Contents

Chapter 1

Introduction to Bayes

1.1 Introduction

Definition 1.1

- A **statistical model** is a mathematical model that embodies a set of statistical assumptions concerning the generation of sample data.
- A statistical model is thus a collection of the probability distribution on some sample space.
- In mathematical terms, a statistical model can be thought of as a triplet $(X, \mathcal{A}, \mathcal{P})$, where X is the set of possible observations (i.e the sample space), $\mathcal{P}$ is the set of probability distribution on X and $\mathcal{A}$ is $\sigma-$algebra of X.
- In a parametric model, we assume that the collection $\mathcal{P}$ is indexed by some set θ.
- The set Θ is called the **parameter set** or, more commonly, the **parameter space**.
- For each $\theta \in \Theta$, let $\mathcal{P}_\theta$ denotes the corresponding member of the collection.
- A parametric statistic model is the triplet

$$(X, \mathcal{A}, \{\mathcal{P}_\theta, \theta \in \Theta\})$$

  where $\Theta \subset \mathbb{R}^d$, d is parameter dimension (number of the scalar parameter in the model).

$\square$

Example 1.1.1

1. **Bernoulli model**

$$(\{0,1\}\,,\ \mathcal{A},\ \{B(p),\ p \in (0,1)\})\,,$$

where $\mathcal{A}$ is the discrete σ-algebra.

2. **Gaussian model**

$$\left(\mathbb{R},\ \mathcal{B}(\mathbb{R}),\ \left\{\mathcal{N}(\mu,\sigma^2) : (\mu,\sigma^2) \in \mathbb{R} \times \mathbb{R}_*^+\right\}\right),$$

3. **Exponential model**

$$\left(\mathbb{R}^+,\ \mathcal{B}(\mathbb{R}^+),\ \{\mathcal{E}(\lambda) : \lambda > 0\}\right)$$

Remark 1.1.1: Non-parametric model

- One can, for example, be interested in "all the probability density function $\mathbb{R}$
- The parametric θ is the density itself and is infinite-dimensional.
- The model is non-parametric

  1. – Choosing a parametric model means admitting that the **"true probability"** governing the phenomenon.
     – This parameter is unknown, and the statistical problem is to select θ in the best way possible. We say that we try to estimate θ, and this of course from the observations.
  2. If there is no prior opinion about the data or when a parameter.
     – The idea is to make as few assumptions as possible on the underlying probability distribution of the observations, to leave it essentially free: the number of parameters may not be fixed and may grow with the amount of data.
     – The set is an infinite dimensional space, e.g, a functional space: the space of all densities, the space of all cumulative distribution functions, and so on.
     – The objective is thus to estimate a function.
     – The only constraint concerns the smoothness of the target functions.

Definition 1.2: Sampling model

- In general, in order to increase the information available on the phenomenon, we use for example the law of large numbers (LLN), or the central limit theorem (CLT), an experiment often consists in n observations that are independent and identically distributed (i.i.d) from chosen parametric family.
- The model in this case becomes the product model or sampling model and can be

written as

$$\left(X^n, \mathcal{A}^{\otimes}, \{\mathcal{P}_\theta^{\otimes n} : \theta \in \Theta\}\right)$$

□

Remark 1.1.2: Notation

- In the following, we will omit the $\sigma-$algebra when defining a statistical model. We will only consider the pair constituted by the space of observations and the family of the probability distribution.
- The latter will be denoted $\mathcal{P}$ and will indicate either $\{\mathcal{P}_\theta\}$ (simple model) or $\{\mathcal{P}_\theta : \theta \in \Theta\}$ (product model)
- Ingredient of Bayes Inference: prior and likelihood.
- The precise definition of likelihood requires the notion of dominant measures and dominated statistical models.

1.2 Domination in statistical model

Let $(X^n, \mathcal{P})$ be a statistical parametric model, where $X \subset \mathbb{R}^k$ is the space of individual observations and $\theta \subset \mathbb{R}$ the parameter space.

Definition 2.1: Absolute continuity of measure

Updating later... □

Definition 2.2: Dominated model

Updating later... □

Example 1.2.1

1. If $\mathcal{P}_\theta$ is the Poisson distribution with parameter $\theta \in \Theta =]0, +\infty[$, the dominating measure is
$$f(x \mid \theta) = e^{-\theta}\frac{\theta^x}{x!},\ x = 0, 1, \ldots$$
2. If $\mathcal{P}_\theta$ is the uniform distribution on $[0, \theta]$, $\theta \in \Theta =]0, +\infty[$, the dominating measure is the Lebesgue measure on $\mathbb{R}^+$ and the density is
$$f(x \mid \theta) = \frac{1}{\theta} \cdot 1_{[0,\theta]}(x).$$
3.

......

Definition 2.3: Likelihood

The likelihood of the sample is the density of $\mathcal{P}_\theta^{\otimes n}$, w.r.t the dominating measure $\mu^{\otimes n}$ and is

$$L(x \mid \theta) = \prod_{i=1}^{n} f(x_i|\theta),$$

where $x_i \in X, \forall i$.
Note: The likelihood function is the joint density interpreted as a function of the parameter, rather than the random variables. □

Remark 1.2.1

Updating later......

Example 1.2.2

1. Compute the likelihood of a Poisson distribution with parameter θ.

$$L(X \mid \theta) = \prod_{i=1}^{n} e^{-\theta} \frac{\theta^{x_i}}{x_i!} = e^{-n\theta} \frac{\theta^{\sum_{i=1}^{n} x_i}}{\prod_{i=1}^{n} x_i!}$$

2. In the case of the uniform distribution on $[0, \theta]$ gives the likelihood

$$L(X \mid \theta) = \prod_{i=1}^{n} \frac{1}{\theta} \cdot 1_{[0,\theta]}(x_i) = \frac{1}{\theta^n} \cdot 1_{[0,\theta]}(x_{(n)}),$$

where $x_{(n)} = \max\{x_1, \ldots, x_n\}$.

3. Normal distribution gives the likelihood

$$L(X \mid \theta) =$$

1.3 Parametric and non-parametric problem

1.3.1 Parametric problem

- In the parametric approach, f is assumed to belong to some parametric family, i.e, $f = f(X \mid \theta)$, where $\theta \in \Theta \subset \mathbb{R}^k$ is unknown.

- We can, for example, look for this density in the family of Gaussian distribution $\{\mathcal{N}(\mu, \sigma^2) : (\mu, \sigma^2) \in \mathbb{R} \times \mathbb{R}_*^+\}$, and in this case, the problem of estimating f comes back to the estimation of a mean and variance.

1.3.2 Non-parametric problem

- When one has no idea about the particular form that the density f can take constructing an estimator of f does not only consist in estimating a mean and a variance as it is the case for Gaussian distribution.

1.4 Bayesian Approach

Theorem 4.1: Bayesian formula

- Let A be an event of interest occurring under assumptions H_i, $i = \overline{1,n}$ with known conditional probability $\mathbb{P}(A \mid H_i)$.
- Let us suppose that the probabilities of the hypothesis $H_1, \dots, H_n$ are also known (prior probability).
- The conditional probabilities (posterior probability of the hypothesis H_i given that A occurred) is

$$\mathbb{P}(H_i \mid A) = \frac{\mathbb{P}(A \mid H_i)\mathbb{P}(H_i)}{\mathbb{P}(A)}$$

with $A = \sum_{i=1}^{n} \mathbb{P}(A \mid H_i)\mathbb{P}(H_i)$.

Definition 4.1: Bayes paradigm

- Let us assume that the observation supporting the statistical analysis $x_1, \dots, x_n$ have been generated from this parameter.
- X_1 has a distribution with a density $f_1(x_1 \mid \theta_1)$, and for all $i = \overline{2,n}$, $X_i \mid X_1, \dots X_{i-1}$ has a density $f_i(x_i \mid \theta_i, x_1, \dots x_{i-1})$ on $\mathbb{R}^p$ where θ_i is known and the function f_i is known.
- We are interested in the joint density

$$X \sim f(x \mid \theta) = f_1(x_1 \mid \theta_1) \prod_{i=2}^{n} f_i(x_i \mid \theta_i, x_1, \dots x_{i-1}) \tag{1}$$

  where $x = (x_1, \dots, x_n)$ is the vector of observations, and $\theta = (\theta_1, \dots, \theta_n)$ is the parameter with components being eventually all equal.
- The model (1) is unifying, in the sense that it represents similarly an isolated observation, dependent observation, and repeated i.i.d observations from a common distribution $f(x_i \mid \theta)$.

- In the latter case of a n-sample, one has

$$f(x \mid \theta) = \prod_{i=1}^{n} f(x_i \mid \theta_i). \tag{2}$$

- Bayesian statistic analysis aims at exploiting in the most efficient way information brought by the X data on the parameter θ, in order to build inference procedures on θ.

- The information provided by the observation x is contained in the density $f(x \mid \theta)$, viewed as a likelihood function of the parameter θ:

$$L(\theta) = L(\theta \mid x) = L_X(\theta) = f(x \mid \theta)$$

□

Remark 1.4.1

- When observing a random phenomenon directed by a parameter θ, statistical methods allow us to deduce from these observations an inference (that is, a summary, a characterization) about θ, while probabilistic modeling characterizes the behavior of the future observations conditional on θ.

- This inverting aspect of Statistics is obvious in the notion of the likelihood function, since, formally, it is just the sample density rewritten in the proper order, i.e, as a function of θ, which is unknown, depending on the observed value x.

Definition 4.2: Prior distribution

- The link between probability properties and Bayesian inference is that, in the Bayesian paradigm, the unknown parameter θ is no longer considered deterministic but is seen as a random variable.

- The parameter space Θ is endowed with the distribution π, which we will call prior distribution, we denote $\theta \sim \pi$.

- This means that we assume that θ is generated according to π, $\theta \sim \pi(\theta)$, "before" x is generated according to $f(x \mid \theta)$

$$\begin{aligned} &X \mid \theta \sim f(x \mid \theta) \\ &\theta \sim \pi \end{aligned}$$

- The key role of the prior distribution in Bayesian statistical analysis lies in the fact that using a prior distribution and its accompanying probabilistic machinery is the most efficient way to summarize the information (on each of information) available about the unknown parameter θ.

□

Remark 1.4.2: abuse of notation

- It is worth mentioning that θ designates both a random variable and a realization.
- Only the context allows the removal of ambiguity.

Definition 4.3

- A (parametric) Bayesian statistical model is made of a parametric statistical model $(X, \mathcal{A}, \{\mathcal{P}_\theta : \theta \in \Theta\})$ and a prior distribution $\pi(\theta)$ on the parameters.
- This definition applies to the product model.

□

Definition 4.4: Posterior distribution

- The posterior distribution is the distribution of θ conditional on X.
- Its density is denoted $\pi(\theta \mid x)$.
- By replacing the joint distribution $f(x \mid \theta)$ with the likelihood $L(\theta \mid x)$, we have

$$\pi(\theta \mid x) = \frac{L_X(\theta)\pi(\theta)}{\int_\theta L_X(\theta)\pi(\theta)},$$

  where $\pi(\theta)$ is the prior, $f(x \mid \theta) = L_X(\theta)$ is the likelihood.
- Inference on θ is based on the posterior distribution.
- The latter plays a primordial role in Bayesian inference and can be interpreted as a summary (in a probabilistic sense) of the information available once X is observed.
- Bayes' theorem somehow updates the prior information on θ (by extracting the information on θ constrained in the observation x) through $\pi(\theta \mid x)$.
- The denomination $\displaystyle\int_\theta f(x \mid \theta)\pi(\theta)d\theta$ is called the marginal distribution of x (marginal likelihood) or predictive, and denoted $m(x)$ or $f(x)$.
- It is used for model comparison and Bayes hypothesis testing.

□

Example 1.4.1: Bayes (1764)

A billiard ball W is rolled on the unit line $[0, 1]$, with a uniform probability of stopping anywhere. It stops at a point p. A second ball O is then rolled n times under the assumptions, and X denotes the number of times the ball O stopped on the left of W. Given $X = x$, what inference can we make on p?

Solution:

Let X_i, $i = \overline{1, n}$ be a random variable such that

$$X_i = \begin{cases} 1 & \text{the ball } O \text{ stops on the left of } W \\ 0 & \text{otherwise} \end{cases}$$

Hence, $X_i \mid p \overset{\text{i.i.d}}{\sim} \text{Ber}(p)$ and then

$$X = \sum_{i=1}^{n} X_i \sim \text{Bin}(n, p).$$

Therefore, the likelihood of x is: $f(x \mid p) = C_n^x p^x (1-p)^{n-x}$
Prior ($p \sim \mathcal{U}([0,1])$): $\pi(p) = 1_{[0,1]}(p)$.
Now, the posterior of p given that x occurred is

$$\begin{aligned} \pi(p \mid x) &= \frac{f(x \mid p)\pi(p)}{\int_0^1 f(x \mid p')\pi(p')dp'} \\ &\propto p^x (1-p)^{n-x} \times 1_{[0,1]}(p) \\ &= p^{x+1-1}(1-p)^{n-x+1-1} \end{aligned}$$

Hence, $p \mid x \sim \text{Beta}(x+1, n-x+1)$.

Remark 1.4.3

If $X \sim \text{Beta}(\alpha, \beta)$, its p.d.f is

$$f_X(x) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{\int_0^1 u^{\alpha-1}(1-u)^{\beta-1}du}$$

and $\displaystyle\int_0^1 u^{\alpha-1}(1-u)^{\beta-1}du = B(\alpha, \beta)$ is the beta function.

In the same spirit, Laplace introduced probabilistic modeling of the parameter space.

Example 1.4.2

- An urn contains a number N of cards (black and white). A card is drawn out of an urn and it is white. We denote p the (unknown) proportion of white cards in the urn. What is the probability that p is equal to p_0?
- In his resolution of the problem, Laplace assumes that all numbers from 2 to $N-1$ are equally likely values for pN, the number of white cards, i.e, that p is uniformly distributed on $\left\{\dfrac{2}{N}, \ldots, \dfrac{N-1}{N}\right\}$.

Solution:
Let X be the event "a white card is drawn". We need to compute

$$\mathbb{P}(p = p_0 \mid x) = \frac{p_0}{\sum_{i=2/N}^{(N-1)/N} p}$$

Let us define $X = \begin{cases} 1, & \text{if it is } W \\ 0, & \text{if it is } B \end{cases}$.
Let us compute the denominator, we have

$$\sum_{p=2/N}^{(N-1)/N} p = \frac{2}{N} + \frac{3}{N} + \ldots + \frac{N-1}{N} = \frac{1}{N}(2 + 3 + \ldots + N - 1)$$
$$= \frac{(N-2)(N+1)}{2N}$$

The posterior is then

$$\mathbb{P}(p = p_0 \mid x) = \frac{2Np_0}{N^2 - N - 2}$$

Example 1.4.3

- Let us consider a n-sample $X = (X_1, \ldots, X_n)$ following an exponential distribution with parameter $\lambda > 0$.
- We are interested in the inverse of this parameter. Let $\theta = \dfrac{1}{\lambda}$.
- We consider as prior on θ an inverse-gamma distribution.
- **Recall:** If $U \sim IG(\alpha, \beta)$, with $\alpha > 0$, $\beta > 0$, then its p.d.f is

$$f(u \mid \alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} \left(\frac{1}{u}\right)^{\alpha+1} e^{-\beta/u}, \ \forall u > 0.$$

Compute the joint distribution of X and θ, and the predictive $m(x)$. From this, deduce the posterior $\pi(\theta|x)$ and determine the nature of this distribution.

- **Indication:** to compute the predictive, you can use Euler's Gamma function defined as follow:

$$\Gamma(z) = \int_0^{+\infty} t^{z-1} e^{-t} dt,$$

where z is complex number such that $\mathrm{Re}(z) > 0$.

Solution:
The joint distribution of $X = (X_1, \ldots, X_n)$ and θ is

$$f(X_1, \ldots, X_n, \theta) = \pi(\theta) \prod_{i=1}^n f(x_i \mid \theta) = \frac{\beta^\alpha}{\Gamma(\alpha)} \theta^{-\alpha-1} e^{-\beta/\theta} \prod_{i=1}^n \frac{1}{\theta} e^{-x_i/\theta} \cdot 1_{[0,+\infty)}(x_i)$$
$$= \frac{\beta^\alpha}{\Gamma(\alpha)} \theta^{-\alpha-n-1} \exp\left\{-\left(\beta + \sum_{i=1}^n x_i\right)\frac{1}{\theta}\right\} \cdot 1_{[0,+\infty)}(x_{(1)})$$
$$= \frac{\beta^\alpha}{\Gamma(\alpha)} \theta^{-\alpha-n-1} \exp\left\{-\left(\beta + \sum_{i=1}^n x_i\right)\frac{1}{\theta}\right\}. \qquad (1)$$

The predictive $m(x)$ is now defined as follow

$$\begin{aligned} m(x) &= \int_0^{+\infty} f(x,\theta)d\theta \\ &= \frac{\beta^\alpha}{\Gamma(\alpha)} \int_0^{+\infty} \left(\frac{1}{\theta}\right)^{\alpha+n+1} \exp\left\{-\left(\beta + \sum_{i=1}^n x_i\right)\frac{1}{\theta}\right\} d\theta \end{aligned}$$

Let $t = \left(\beta + \sum_{i=1}^n x_i\right)\frac{1}{\theta}$, hence $dt = \left(\beta + \sum_{i=1}^n x_i\right)\frac{-1}{\theta^2}d\theta$. So, we obtain now that

$$\begin{aligned} m(x) &= \frac{\beta^\alpha}{\Gamma(\alpha)} \int_{+\infty}^0 \left(\frac{t}{\beta + \sum_{i=1}^n x_i}\right)^{\alpha+n+1} e^{-t} \left(\beta + \sum_{i=1}^n x_i\right) \frac{dt}{-t^2} \\ &= \frac{\beta^\alpha}{\Gamma(\alpha)\left(\beta + \sum_{i=1}^n x_i\right)^{\alpha+n}} \int_0^{+\infty} t^{\alpha+n-1} e^{-t} dt \\ &= \frac{\beta^\alpha \Gamma(\alpha+n)}{\Gamma(\alpha)\left(\beta + \sum_{i=1}^n x_i\right)^{\alpha+n}} \end{aligned}$$

From (1), we can deduce the posterior

$$\begin{aligned} \pi(\theta \mid x) &\propto f(x_1, \ldots, x_n, \theta) \\ &\propto \theta^{-\alpha-n-1} \exp\left\{-\left(\beta + \sum_{i=1}^n x_i\right)\frac{1}{\theta}\right\}, \end{aligned}$$

which implies that $\theta \mid x \sim \text{IG}\left(\alpha + n,\ \beta + \sum_{i=1}^n x_i\right)$.

1.5 Point Estimation

- In order to simplify the writing, we consider a rest of parameter θ.
- The results generalize to the higher dimension, the integrals are just more tedious.
- Let us consider the parametric model (with one observation) $(x \in X, \{\mathcal{P}_\theta : \theta \in \Theta\})$, a prior π on θ, and dominant measure μ of the model

$$f(x \mid \theta) = \frac{d\mathcal{P}_\theta}{d\mu}$$

- **Recall.** The posterior is

$$\pi(\theta \mid x) = \frac{f(x \mid \theta)\pi(\theta)}{\int_\theta f(x \mid \theta)\pi(\theta)d\theta},$$

where $\int_\theta f(x \mid \theta)\pi(\theta)d\theta = m(x)$.

1.6 Introduction to Decision Theory

1.6.1 Evaluating estimators

- Considering that the overall purpose of most inferential studies is to provide the statistician (or a client) with a decision, it seems reasonable to ask for an evaluation criterion of decision procedures.
- In order to compare procedures, we need to calculate which procedure is the best even though we cannot observe the nature of the parameter space Θ.
- The criterion is usually called loss.
- This must assess the consequence of each decision, and depend on the parameter of the model, i.e, the true state of the world (or of nature).
- **Loss function** are functions of the state of the Nature θ and a decision function $\delta(\cdot)$.
- This is the main challenge of decision theory and the break between frequentists and Bayesian.
- Let D denote the set of the possible decision. It is called the decision space,

Definition 6.1: Loss function

A loss function is any function $L : \Theta \times D \to \mathbb{R}^+$ of the state of the Nature θ and a decision function $\delta(\cdot)$. $\square$

- The main requirement for a good point estimator is that the error $(\hat{\theta}_n - \theta)$ is closed to 0.
- To determine a point estimator of θ, one can compare the approximation $\delta(x)$ of θ, by means of a loss function, $L(\theta, \delta(x))$, which quantifies the consequence of the error made by replacing the "true" parameter θ by its approximation $\delta(x)$.
- Typically, the greater distance between $\delta(x)$ and θ, the greater value of $L(\theta, \delta(x))$.
- A classically used loss function is the **quadratic loss function**, which is defined as

$$L(\theta, \delta(x)) = (\theta - \delta(x))^2.$$

- A variant of this function is the weighted quadratic cost function

$$L(\theta, \delta(x)) = w(\theta)(\theta - \delta(x))^2.$$

1.6.2 Decision theory

- From a decision-theoretic point of view, the statistical model now involves three spaces: observation space X, parameter space Θ, and decision space D (a.k.a action space).
- Statistical inference then consists of taking a decision, denoted $d \in D$ (or a, or $\delta(x)$) and related to the parameter $\theta \in \Theta$, based on the observation $x \in X$; x and θ related by the distribution $f(x \mid \theta)$.

- In most cases, the decision d will be to evaluate (estimate) a function of θ, as accurately as possible.
- **Example:** Estimating a mean by the sample mean $\delta(x) = \overline{X_n}$.
- **Decision theory** assumes in addition that each action can be evaluated via the loss function.

1.6.3 Frequentist risk

- A decision is "good" is good if it leads to a zero loss.
- In other words, a good decision is the solution to the equation

$$L(\theta, \delta(x)) = 0.$$

- Since θ is unknown, we obviously cannot solve this equation.
- One could think of minimizing the loss function $L(\theta, \delta(x))$ in $\delta(x)$, but this minimization is often impossible.
- In order to derive an effective comparison criterion from the loss function, the frequentist approach proposes to consider instead the average loss (named frequentist risk).

Definition 6.2

The frequentist risk is the mean risk (expectation) of the loss function

$$\begin{aligned} R(\theta, \delta) &= \mathbb{E}_\theta\left[L(\theta, \delta(x))\right] \\ &:= \int_X L(\theta, \delta(x)) d\mathcal{P}_\theta(x) \\ &= \int_X L(\theta, \delta(x)) f(x \mid \theta) d\mu(x) \end{aligned}$$

for a μ-dominated model. □

Example 1.6.1

Using the quadratic loss $L(\theta, \delta(x)) = (\theta - \delta(x))^2$, the frequentist risks is

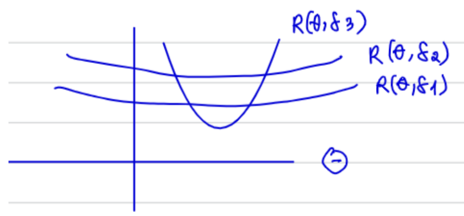
$$\begin{aligned} R(\theta, \delta) &= \mathbb{E}_\theta[L(\theta, \delta(x))] = \mathbb{E}_\theta[(\theta - \delta(x))^2] \\ &= \mathbb{E}_\theta[(\theta - \mathbb{E}_\theta[\delta(x)] + \mathbb{E}_\theta[\delta(x)] - \delta(x))^2] \\ &= \mathbb{E}_\theta[(\theta - \mathbb{E}_\theta[\delta(x)])^2] + \mathbb{E}_\theta[(\mathbb{E}_\theta[\delta(x)] - \delta(x))^2] - 2(\theta - \mathbb{E}_\theta[\delta(x)])\mathbb{E}_\theta[\mathbb{E}_\theta[\delta(x)] - \delta(x)] \\ &= (\theta - \mathbb{E}_\theta[\delta(x)])^2 + \mathbb{E}_\theta[(\mathbb{E}_\theta[\delta(x)] - \delta(x))^2] \end{aligned}$$

1. The bias of $\delta(x)$ is $b_\theta[\delta(x)] = \mathbb{E}_\theta[\delta(x)] - \theta$.
2. Its variance is $\mathrm{var}_\theta[\delta(x)] = \mathbb{E}_\theta[(\mathbb{E}_\theta[\delta(x)] - \delta(x))^2]$

- Hence, we have

$$R(\theta, \delta) = b_\theta^2[\delta(x)] + \text{var}_\theta[\delta(x)]$$

- This result allows frequentists to analyze the variance and the bias respectively and can be used to motivate frequentist ideas (e.g minimum variance unbiased estimators).
- The frequentist paradigm relies on this criterion to compare estimators and, if possible, to select the best estimator.
- Notice, however, that there are several difficulties associated with this approach.
    1. The error (loss) is averaged over the different values of X proportionally to the density $f(x \mid \theta)$. The observation X is not taken into account any further. Such evaluation may be satisfactory for the statistician, but it is not so appealing for a client, who wants optimal results for his data X, not that of another!
    2. For a procedure δ, the risk $R(\theta, \delta)$ is a function of the parameter θ. Therefore, the frequentist approach does not induce a total ordering on the set of procedures.
    $\longrightarrow$ It is generally impossible to compare decision procedures with this criterion since two crossing risk functions prevent comparison between the corresponding estimators.



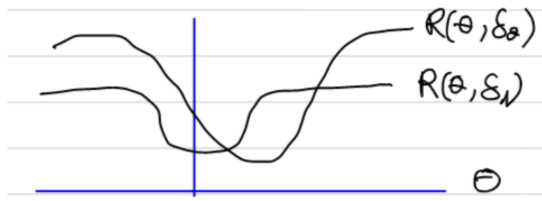
- Which decision is better between δ_1, δ_2 and δ_3?
- Often one decision does not dominate the other everywhere as in the case with decisions δ_1 and δ_2.
- The challenge is in saying whether, for example, δ_1 or δ_3 is better.
- Frequentists have a few answers for deciding which is better.
Q1: Admissibility
    - A decision that is inadmissible is one that dominated everywhere. Example: δ_2 compare to δ_1
    - It would be easy to compare decisions if all one were inadmissible. But usually, they overlap, so this criterion fails.

Q2: Restricted classes of procedures
    - The best procedure can be only obtained by restricting the set of authorized procedures.
    - For example, if we restricting our class of procedures to just those which are unbiased procedures is a nice frequentist theory, but many good procedures are biased.
    - More surprisingly, some unbiased procedures are actually inadmissible.

- **Example:** as shown by Stein, the sample mean is an inadmissible estimate of the mean of a multivariate Bayesian in 3D.

Q3: Minimaxity

- The minimax criterion appears as insurance against the worst case because it aims at minimizing the expected loss in the least favorable case.
- In this approach, we get around the problem by just looking at $\sup_{\theta} R(\theta, \delta)$.
- **Example:**



Which decision is better in a minimax sense?
- **Recall.** The frequentist risk

$$R(\theta, \delta) = \mathbb{E}_{\theta}[L(\theta, \delta)] = \int_X L\left[(\theta, \delta(x))\right] d\mathcal{P}_{\theta}(x).$$

Note again that it is an expectation on X assuming fixed θ.
- A Bayesian would only look at x, the data you observed, not all possible X.
- However, the frequentist insists on avoiding priors so that they can derive simple procedures that can be used everywhere, i.e, software that can be used on multiple datasets. This is valid.
- An alternative definition of a frequentist is "someone who is happy to look at other data that he could have obtained but did not".

1.6.4 Bayes Decision Theory

- It relies on the posterior expected loss, called the posterior risk, based on the data X and a prior π:

$$p(\pi, \delta \mid x) = \mathbb{E}^{\pi(\cdot|x)}\left[L(\theta, \delta(x))\right] = \mathbb{E}^{\pi}\left[L(\theta, \delta(x)) \mid x\right] = \int_{\theta} L(\theta, \delta(x))\pi(\theta \mid x) d\theta.$$

- We average the loss according to the posterior distribution of the parameter θ, conditionally on the observed value x.
- The posterior expected loss is thus a function of x.
- Note that the prior enters the equation when calculating the posterior density.
- Using the Bayes risk, we can define a bit of Jargon.
- The Bayes risk is

$$r^{\pi} = r(\pi, \delta^{\pi}).$$

This is just $r(\pi, \delta)$ with the Bayes action δ^{π} plugged in and it is the **integrated risk** defined below.

Definition 6.3: Integrated risk

Given a prior distribution, it is also possible to define the integrated risk, which is the frequentist risk averaged over the values of θ according to their prior distribution

$$r(\pi,\delta) := \mathbb{E}^{\pi}\left[R(\theta,\delta)\right] = \int_{\theta} R(\theta,\delta)\pi(\theta)d\theta$$
$$= \int_{\theta}\int_{X} L(\theta,\delta(x))f(x\mid\theta)d(x)\pi(\theta)d\theta.$$

□

- One particular interest of this second concept is that it associates a real number with every estimator.
- It therefore induces a total ordering on the set of estimators.
- This implies that, while taking into account the prior information through the prior, the Bayesian approach is sufficiently reductive (in a positive sense) to reach an effective decision.
- Moreover, the above two notions (integrated and posterior risks) are equivalent in that they lead to the same decision.
- A decision minimizing the integrated risk $r(\pi,\delta)$ can be obtained by selecting, for every $x \in X$, the value $\delta(x)$ which minimizes the posterior expected loss $p(\pi,\delta \mid x)$.
- This result from the following proposition:

Proposition 1.6.1

The integrated risk $r(\pi,\delta)$ is the average (mean) of the posterior risk $p(\pi,\delta \mid x)$ w.r.t the marginal distribution $m(x)$.

Proof.

Proof:

$$r(\pi,\delta) = \int_{\Theta}\left(\int_{X} L(\theta,\delta(x))\, f(x|\theta)dx\right)\pi(\theta)\,d\theta$$

$$= \int_{X}\left(\int_{\Theta} L(\theta,\delta(x))\, f(x|\theta)\,\pi(\theta)d\theta\right)dx \quad \text{(Fubini - Tonelli)}$$

$$= \int_{X}\left(\int_{\Theta} L(\theta,\delta(x))\,\pi(\theta|x)d\theta\right) m(x)\,dx$$

(using that $f(x|\theta)\,\pi(\theta) = m(x)\,\pi(\theta|x)$ and we take out $m(x)$ when integrating w.r.t θ)

$$= \int_{X} p(\pi,\delta|x)\, m(x)\,dx \quad \text{(the inside integral being } p(\pi,\delta|x)\text{)}$$

This result leads to the following definition 2 of a Bayes estimator

Definition 6.4: Bayes estimator

- A Bayes estimator associated with a prior distribution π and a loss function L is any estimator δ^π which minimizes $r(\pi, \delta)$.
- For every $x \in X$, it is given by $\delta^\pi(x)$, argument of $\min_{\delta \in D} p(\pi, \delta \mid x)$

$\square$

Proposition 1.6.2

Under the squared-norm loss the Bayes action (optimal decision) when estimating θ is to choose the posterior mean

$$\delta^\pi(x) = \mathbb{E}^{\pi(\cdot|x)}[\theta]$$

Proof.
Back to the previous example.

- In this example, we have obtained a posterior distribution, for $\theta = \dfrac{1}{\lambda}$ an inverse-gamma with parameters $\left(n+\alpha, \sum_{i=1}^{n} x_i + \beta\right)$.
- The expectation of an inverse-gamma distribution with parameter (a, b) is $\dfrac{b}{a-1}$.
- The Bayes estimator of θ, using the squared-error loss is

$$\hat{\theta} = \delta^\pi(x) = \frac{\sum_{i=1}^{n} X_i + \beta}{n + \alpha - 1}$$

Definition 6.5: The absolute error loss

An alternative to the quadratic loss in one dimension is to use the absolute error loss:

$$L(\theta, \delta(x)) = |\theta - \delta(x)|, \tag{3}$$

or more generally, a multilinear function

$$L(\theta, \delta(x)) = \begin{cases} k_2(\theta - \delta(x)) & \text{if } \theta > \delta(x) \\ k_1(\theta - \delta(x) & \text{otherwise} \end{cases} \tag{4}$$

$\square$

Proposition 1.6.3

A Bayesian estimator associated with the prior distribution π and the multilinear loss (4) is a $\dfrac{k_2}{k_1+k_2}$ fractile of $\pi(\theta \mid x)$.
In particular, if $k_1 = k_2$ in the case of absolute error loss (3), the Bayes estimator is the posterior median.

1.7 Modelling the prior information

How do we choose prior?

- Undoubtedly, the most critical and most criticized point of Bayesian analysis deal with the choice of the prior.
- Choosing the prior is a fundamental step in Bayesian analysis.
- We call prior information on the parameter θ any information available on θ, apart from that brought by the observations.
- The choice of prior should be based on experiences, on an intuition, an idea that the practitioner has of the random phenomenon he is observing.
- However, this choice is quite difficult to make:
  - It can be difficult to summarize information through a single distribution.
    $\longrightarrow$ Many probability distributions may be compatible with the information.
  - Different priors often lead to different conclusions.
- In practice, the choice of priors can be motivated by computational aspects (**choice of conjugate distributions**)
- Or to take into account the fact that we do not know much about the parameter and thus use non-informative priors.
- On the choice of the priors depends on the calculation of the posteriors.
- We can use families of the so-called conjugate priors.
- The main justification for the use of conjugate priors is that they are computationally convenient and they have asymptotically desirable properties.

Definition 7.1: Conjugate priors

A family of priors $\mathcal{F}$ is the conjugate if the prior π upon being multiplied by the likelihood, yields a posterior $\pi(\theta \mid x)$ in the same family. $\square$

Remark 1.7.1: Propositional notation

- We recall that Bayes formula is given by

$$\pi(\theta \mid x) = \frac{L(\theta \mid x)\pi(\theta)}{\int_\theta L(\theta \mid x)\pi(\theta)d\theta}$$

- The notation is often used

$$\pi(\theta \mid x) \propto L(\theta \mid x)\pi(\theta),$$

to say that the posterior is actually proportional to the likelihood, multiplied by the prior distribution.

- In fact, $m(x)$ is seen as a constant in the sense that it does not depend on θ.

1.7.1 Non-informative priors

- Conjugate prior are computationally nice, but objective Bayesians instead prefer priors which do not strongly influnce the posterior distribution
- Such a prior is called non-informative (uninformative)
- It is also called the defauth prior on the objective prior
- This is an approach to use when there is no information, or when one do not wants to influnce the study.
- The historical approach, followed by laplace and Bayes, was to assign flat priors
- These priors seem reasonably uniformative
- We do not know where the actual value lies in the parameter space, so we might as well consider all values equiprobable.

- Example : a binominal distribution $X \sim B(n,p)$
- The flat prior on p is the uniform distribution on the unit interval, $\pi(p)=1$
- The uniform distribution does not favour any particular value and can be seen as a way to represent ignorance
- If you are ignorant about a parameter θ you are likely to be equally ignorant about $\theta^* = h(\theta)$, for some function h (with differentiable inverse)
- But if θ has a uniform prior $\pi_\theta(\theta) = 1$ then with $g = h^{-1}$ this corresponds to cdup this prior on $\theta^* = h(\theta)$

$$\pi_{\theta^*}(\theta^*) = \pi_\theta(g(\theta^*))\,|g'(\theta^*)| = |g'(\theta^*)|$$

(see theorem on tranformation on random variables)

- Hence, ignorace represented by the uniform prior distribution does not translate across scales
- A prior that does translate across scale is Jeffey's prior

1.7.2 Jeffreys' priors

Idea behind Jeffreys' priors

- Suppose we are provided with some model and some data, i.e, with a likelihood function $f(x \mid \theta)$.
- One should be able to manipulate the likelihood and get a prior on θ, from the likelihood only.
- **Note:** This approach goes contrary to the subjective Bayesian frame of mind.

Jeffreys' rule

- The method proposed by Jeffreys (1961) makes it possible to construct non-informative priors.
- These are based on **Fisher Information** $I(\theta)$.
- The argument could be this: $I(\theta)$ is an addition of the amount of information brought by the observation about θ.
- The larger $I(\theta)$ is the more the observation brings information.
- Favoring the values of which $I(\theta)$ is large is equivalent to minimizing the influence of the prior distribution and is therefore as non-informative as possible.
- In the **one-dimensional case**, Jeffreys' rule consists in considering priors of the form:

$$\pi(\theta) \propto \sqrt{I(\theta)},$$

where $I(\theta) = -\mathbb{E}\left[\frac{\partial^2}{\partial\theta^2} \log f(x \mid \theta)\right]$.

- In the **multi-dimensional case**, $\theta = (\theta_1, \ldots, \theta_q)$, one can consider priors of the form

$$\pi(\theta) \propto \sqrt{|\det(I(\theta))|}$$

where $I(\theta)$ is the Fisher information matrix

$$[I(\theta)]_{i,j} = -\mathbb{E}\left[\frac{\partial^2}{\partial\theta_i\partial\theta_j} \log f(x \mid \theta)\right], \ 1 \leq i, j \leq q.$$

Example 1.7.1: Binomial likelihood

Let $X \sim \text{Bin}(n, \theta)$, $\theta \in]0, 1[$ unknown. Compute the Jeffreys prior of θ.

Solution:
Since $X \sim \text{Bin}(n, \theta)$ then the likelihood is

$$f(x \mid \theta) = C_n^x \theta^x (1 - \theta)^{n-x}$$

which implies that

$$\ln f(x \mid \theta) \propto X \ln\theta + (n - x)\ln(1 - \theta).$$

So, we have

$$
\begin{aligned}
&\frac{\partial f}{\partial \theta} \propto \frac{x}{\theta} - \frac{n-x}{1-\theta} \\
\Rightarrow &\frac{\partial^2 f}{\partial \theta^2} \propto \frac{-x}{\theta^2} - \frac{n-x}{(1-\theta)^2}
\end{aligned}
$$

The Fisher information will be

$$
\begin{aligned}
I(\theta) &= -\mathbb{E}\left[\frac{\partial^2 \ln f}{\partial \theta^2}\right] = \mathbb{E}\left[\frac{x}{\theta^2} + \frac{n-x}{(1-\theta)^2}\right] \\
&= \frac{n\theta}{\theta^2} + \frac{n-n\theta}{(1-\theta)^2} = \frac{n}{\theta} + \frac{n}{1-\theta} \\
&= \frac{n}{\theta(1-\theta)}
\end{aligned}
$$

The Jeffreys priors of θ will be

$$
\pi(\theta) \propto \sqrt{I(\theta)} \propto \theta^{-1/2}(1-\theta)^{-1/2},
$$

which implies that $\theta \sim \text{Beta}\left(\frac{1}{2}, \frac{1}{2}\right)$

Example 1.7.2: Normal likelihood

Let $X \sim \mathcal{N}(\mu, \sigma)$ with $\theta = (\mu, \sigma)$. Compute the Jeffreys prior of θ.

Solution.

Example 1.7.3: Normal information

Let $X \sim \mathcal{N}(\mu, \sigma^2)$ with $\theta = (\mu, \sigma^2)$. Compute the Jeffreys prior of θ.

Solution.

Remark 1.7.2

- Jeffreys' priors are widely used in Bayesian Analysis.
- In general, they are not conjugate priors.
- The fact that we ended up with a conjugate Beta prior to the Binomial example is just a lucky confidence.
- For example, with a Gaussian model, we showed that $\pi(\mu) \propto 1$ and $\pi(\sigma) \propto \dfrac{1}{\sigma^2}$, which do not look anything like a Gaussian or an inverse gamma, respectively.
- However, it can be shown that Jeffreys' priors are limits of conjugate prior densities.
- **Examples:**
  1. A Gaussian density $\mathcal{N}(\mu_0, \sigma_0^2)$ approaches a flat of prior as $\sigma_0 \to +\infty$.
  2. The inverse-gamma on $\sigma \propto \sigma^{-(a+1)} e^{-b/\sigma} \to 1/\sigma$ as $a, b \to 0$.
- Instead of the posterior expectation (Bayes estimator under quadratic risk), one can alternatively choose as estimator the maximizing the posterior.

Definition 7.2: Maximum A Posterior (MAP)

We call **maximum a posterior (MAP)** any estimator

$$\delta^\pi(x) \in \arg\max_\theta \pi(\theta \mid x).$$

- This notion is the Bayesian counterpart of the frequentist maximum likelihood estimator.
- Notice that the MAP estimator also maximizes $L(\theta \mid x)$ thus by passing the computation of the marginal.
- Its drawbacks are the same as MLE: non-uniqueness, instability (due to optimization) calculation, etc.

□

Definition 7.3

Let $\hat{\theta}(x)$ be an estimator of θ. Define the loss

$$L_\varepsilon(\theta, \hat{\theta}(x)) = \begin{cases} 0, & \text{if } |\theta - \hat{\theta}(x)| \le \varepsilon, \\ 1, & \text{if } |\theta - \hat{\theta}(x)| > \varepsilon. \end{cases}$$

□

Proposition 1.7.1

When $\varepsilon \to 0$, the MAP estimate is the limit of the Bayesian estimate associated with this loss.

Example 1.7.4

Let $X \sim Bin(n,p)$. Compute the corresponding MAP estimator for each of the two priors:

1. $\pi_1(p) = \dfrac{1}{\text{Beta}(1/2,1/2)} p^{-1/2}(1-p)^{-1/2}$.
2. $\pi_2(p) = 1$.

Solution.
Since $X \sim Bin(n,p)$ then $f(x \mid p) = C_n^x p^x (1-p)^{n-x}$.

1. The posterior of p with known prior π_1 is

$$\begin{aligned}\pi_1(p \mid x) &= f(x \mid p)\pi_1(p)\\ &\propto p^x(1-p)^{n-x}p^{-1/2}(1-p)^{-1/2}\\ &= p^{x-1/2}(1-p)^{n-x-1/2}.\end{aligned}$$

   Hence, $p \mid x \sim \text{Beta}\left(x+\dfrac{1}{2}, n-x+\dfrac{1}{2}\right)$.
   The log posterior now is defined as

$$\begin{aligned}\ln \pi_1(p \mid x) &\propto \left(x-\frac{1}{2}\right)\ln p + \left(n-x-\frac{1}{2}\right)\ln(1-p)\\ \Rightarrow \frac{\partial \ln \pi_1(p \mid x)}{\partial p} &= \frac{x-1/2}{p} - \frac{n-x-1/2}{1-p}\\ \frac{\partial \ln \pi_1(p \mid x)}{\partial p} &= 0 \Leftrightarrow p = \frac{x-1/2}{n-1}.\end{aligned}$$

   (Have to check $\dfrac{\partial^2 \ln \pi(p \mid x)}{\partial p^2} < 0$)
   Therefore, the MAP is $\hat{p}_{\text{MAP}} = \dfrac{x-1/2}{n-1}$.

2.

1.7.3 Confidence Intervals (frequentist approach)

- We are looking for a **confidence interval for an unknown proportion** p.
- The estimated proportion on a sample of size 1000 is 0.53 and the estimated CI is $[0.5, 0.56]$, for a confidence level of 95%.

How do you interpret this result?

Remark 1.7.3

The formula of an estimated confidence interval (CI), at level 95%, for a proportion p (n large)

$$\left[f - \frac{\sqrt{f(1-f)}}{\sqrt{n}} \times 1.96, f + \frac{\sqrt{f(1-f)}}{\sqrt{n}} \times 1.96\right],$$

- where f is the estimated value of p on the sample (here 0.53, the sample size n (here 1000) and 1.96 the quantile of order 97.5% of $\mathcal{N}(0,1)$ and the risk is $\alpha = 5\%$.
- We recall that the bounds of the confidence interval for a parameter θ are random quantities that vary from one sample to another.
- The correct interpretation of the confidence interval at level 95% is then the following: "95% of the intervals calculated on the whole of the possible samples (all those which it is possible to draw) contain the true value θ'.
- Frequentists assume that there exists a singular true value of the unknown parameter.
- The possible values of the parameter are not probabilized.
- Additionally, for a given set of data, the confidence interval is fixed.
- In the previous example, the bounds obtained for the observed sample are [0.5;0.56]
- The event "$0.50 \leq p \leq 0.56$ " is true or false since p is fixed at and we cannot assign any probability to it, otherwise 1 or 0 if the true value is or is not contained by this interval (a probability), of course, that we can never observe)
- It is thus illegitimate to write $P(0.5 \leqslant p \leqslant 0.56) = 0.95$ or to state that: "There is a 95%. chance that the unknown parameter p is between 0.5 and 0.56".
- Frequentist confidence intervals are based upon the notion of repeated trials and precise interpretation and description of these intervals is unintuitive and cumbersome.
- **Example**: intervals constructed in this manner are expected to contain the true parameter 95% of the time.
- This interpretation doesn't answers the actual question of interest: "What is the probability that the parameter estimated from a given data set lies in this range?".
- Bayes confidence intervals allow this natural interpretation.

1.7.4 Bayesian Credible Intervals (BCI)

- In bayesian inference, on the contrary, we probabilities the possible values of the parameter.
- From an initial state of knowledge formalized by the prion, and the data, we obtain a posterior distribution that directly expresses our uncertainty on the parameter, conditional on the particular sample observed.

- From the posterior, we can deduce arrange for the parameter, usually called (Bayesian) credible interval to distinguish it from the confidence interval.

Definition 7.4: α-credible

A set C of Θ is α-credible iff

$$\begin{aligned}\mathbb{P}^{\pi|x}(\theta \in C) &:= \mathbb{P}^{\pi}(\theta \in C \mid x) = \mathbb{E}^{\theta|x} \cdot 1_{[\theta \in C]} \\ &= \int_C \pi(\theta \mid x) d\theta \geq 1 - \alpha\end{aligned}$$

$\square$

- Given a posterior distribution, such a region is not unique (there is an infinity of them)

- We will be interested in the one of minimal volume.

- **Idea:** to minimize their volume, these sets must contain the most likely regions of the posterior distribution, hence the name "Highest Posterior Density".

Definition 7.5: The highest posterior

The highest posterior density (HPD)-region at level $(1-\alpha)100°\%$ for θ, is a subset $c \subset \oplus$ such that

$$C = \{\theta \in \Theta : \pi(\theta \mid x) \geqslant k(\alpha)\},$$

Where $k(\alpha)$ is the largest bound such that.

$$\mathbb{P}^{\pi/x}(\theta \in C) = \int_C \pi(\theta \mid x) d\theta = 1 - \alpha$$

$\square$

Example 1.7.5

- We have seen (g Tutorial 2) that if $X \sim \mathcal{N}(\theta, 1)$ and $\theta \sim \mathcal{N}(0, 10)$ then $\theta \mid x \sim \mathcal{N}\left(\frac{10}{11}x, \frac{10}{11}\right)$
- Let's build a credible interval (HPD region in 1D) for θ.
- We have
$$Z := \frac{\theta - \frac{10}{11}x}{\sqrt{\frac{10}{11}}} \sim \mathcal{N}(0, 1)$$
Then $P\left(Z \in \left[-z_{\alpha/2}; z_{\alpha/2}\right]\right) = 1 - \alpha$ where $z_{\alpha/2}$ is the $\alpha/2$-quantile of $\mathcal{N}(0, 1)$.
- The interval
$$C_\alpha = \left[\frac{10}{11}x - z_{\alpha/2}\sqrt{\frac{10}{11}}; \frac{10}{11}x + z_{\alpha/2}\sqrt{\frac{10}{11}}\right]$$
has a posterior probability of $(1 - \alpha)$ to contain θ.

Example 1.7.6: IQ test

Jeremy is an enthusiastic student, and poses a statistical model for his scores on standard IQ tests. He thinks that, in general, his scores are normally distributed with unknown μ and variance $\sigma^2 = 80$ Prior (and expert) opinion is such that μ is a normal random variable distributed as follow, $\mu \sim \mathcal{N}\left(\mu_0 = 110, \sigma_0^2 = 120\right)$. Jeremy tools the test and scored 98

1. Give frequentist and Bayesian estimates of μ.
2. Give the CI and BCI for μ at level 95%.
3. Give BCI when $\sigma_0^2 = 10000$. Comment on the result.

- Recall: two-sided CI formula for μ (if σ is known):
$$\left[\bar{X}_n - \frac{\sigma}{\sqrt{n}}\phi^{-1}(1 - \alpha/2); \overline{X_n} + \frac{\sigma}{\sqrt{n}}\phi^{-1}(1 - \alpha/2)\right]$$
- Recall: If $x \sim \rho^P\left(\mu, \sigma^2\right)$ with $\theta = \mu$ and σ^2 known. If we assume μ with Gaussian prior, $\mu \sim dP\left(\mu_0, \sigma_0^2\right)$. Then the posterior $(n = 1)$:
$$\mu \mid x \sim \mathcal{N}(\mu_n, \sigma_n^2)$$
where $\mu_n = \left(\frac{\sigma_0^2}{\sigma_0^2 + \sigma_n^2}\right) x + \frac{\sigma^2}{\sigma^2 + \sigma_0^2}\mu_0$, and $\sigma_n^2 = \frac{\sigma^2}{\sigma^2 + \sigma_0^2}\sigma_0^2$.

Solution.

1. Here $\sigma^2 = 80, \mu_0 = 110, \sigma_0^2 = 120$ and $x = 98$.
The posterior of μ is $\mu \mid x \sim \mathcal{N}(102.8, 48)$.
The frequentist estimate of μ is $\hat{\mu}_{\text{est}} = 98$ (it's $\bar{x}_n$)

2. A 95% CI is

$$[98 - 1.96\sqrt{80},\ 98 + 1.96\sqrt{80}] = [80.46923,\ 115.5308]$$

- Hence, the length of this interval is approximately 35.
- Let's move to the Bayesian counter pants.
- The Bayes estimate is $\pi_n := \delta''(x) = \mu_n = 102.8$.
- The BCI at level 95% is given by the formula

$$\left[\mu_n - \sigma_n \phi^{-1}(1-\alpha/2),\ \mu_n + \sigma_n \phi^{-1}(1-\alpha/2)\right]$$

- By applying this, we get

$$[102.8 - 1.96\sqrt{48},\ 102.8 + 1.96\sqrt{48}] = [89.22072,\ 116.3793].$$

  Therefore, the BCI has a length of approximately 27.
- Note that the BCI is shorter than the CI.
- This is a consequence of the fact that the posterior variance is smaller than that of the likelihood, thanks to the incorporation of prior information.
- Given the observations, we have a 0.95 probability that the true value of the parameter lies between the limits of the BCI.
- The Bayesian framework provides a legitimate interpretation of the confidence intervals in the terms of probabilities on parameters.

3. - For $\sigma_0^2 = 10000, \mu_n = 98.0952$ and $\sigma_n^2 = 79.3651$.
   - The BCI is

$$[80.6342,\ 115.5563]$$

   with a length of 34.9221.
   - The BCI in this case is almost identical to the CI.

Remark 1.7.4

- The BCI depends on the prior distribution.
- When the prior is highly spread out and reflects initial information that is relatively vague compared to the information provided by the data, the BCI has the same bounds as the CI.
- When, on the other hand, the distribution is not very spread out and reflects relatively precise initial information, this has the consequence of reducing the dispersion of this distribution (because we have more information), and the BCI is then more precise.

Chapter 2

MCMC method

2.1 Why MCMC method?

- Let us consider, for example, the simplified relationship

$$y = Hx,$$

where H is a linear operator, $y = y_1, \ldots, y_M$ the observation (model outputs) and $x = x_1, \ldots, x_N$ the unknown (model inputs).

- The purpose of inversion is to find x from y (knowing H).

- In Bayesian statistics, we assume a distribution on the model parameters (here the vector x).

- Once the posterior distribution is known, we compute an estimator for x (maximum a posterior, posterior mean, etc).

- The maximum a posterior (MAP) estimator is

$$\hat{x}_{MAP} = \arg\max_x \pi(x \mid y)$$

- The posterior mean (PM) estimator is given by

$$\hat{x}_{PM} = \mathbb{E}(x \mid y) = \int x\pi(x \mid y)dx.$$

- MAP and PM estimators require respectively optimization and integration techniques.

- However, this comes out with some practical problems.

- The high dimensionality of the variables considered (pixels of an image, samples or a signal, etc) makes integration and optimization calculations very difficult or impossible.

- As soon as the dimension of the integration space is greater than or equal to three, the numerical calculation is delicate and simulation methods are used.

- An attempted solution to estimating the mean would be to generate a sample $X^{(1)}, \dots, X^{(k)}$ and to approximate the integral by the empirical mean:

$$\hat{X}_{PM} \approx \frac{1}{k} \sum_{i=1}^{k} X^{(i)},$$

when k is large enough.

- This assumes, however, that we can generate samples according to the posterior distribution.
- We cannot use classical simulation techniques such as inverse transform sampling (inversion of the CDF, to generate samples when a CDF is known and invertible CDF). Box-Muller transforms, etc, in the multidimensional case (except....)

2.2 MCMC method

2.2.1 Problem matic

- We want to generate samples according to a target distribution of interest that is not known or difficult to simulate.
- In the Bayes framework, it is the posterior distribution often known up to up a multiplicative constant.
- The MCMC algorithm builds a transition kernel whose invariant distribution is the target distribution.
- It is then necessary to ensure the convergence of the Markov chain produced by the algorithm towards the target distribution.
- To do so, one has to check irreducibility, recurrence, and aperiodicity conditions on the kernel.
- We introduce two major techniques devised to create Markov chains with a given distribution, namely Metropolis-Hastings algorithms and Gibb Sampling

2.3 Metropolis-Hastings (MH) Algorithm

2.3.1 Principle

- The Metropolis-Hastings algorithm is a simulation scheme that allows generating samples according to a probability density $\pi(x)$ known up to a normalizing constant (that is independent of X).
- This may be the case of posteriors which are always known up to a proportionality constant as products of the likelihood and a prior.
- The MH algorithm uses a principle of acceptance and rejection of variables generated by a conditional density $q(\cdot \mid x)$.
- The distribution $\pi(x)$ is often called target or objective distribution, whereas the conditional $q(\cdot \mid x)$ is called instrumental or proposal, or candidate distribution.
- This simulation scheme is presented in the table below.

2.3.2 Sampling scheme of the MH algorithm

Sampling scheme of the MH algorithm

1. Iteration $t = 0$: start with an arbitrary x_0 from the support of target distribution.

2. Iteration $t + 1$:

   - Given $X^{(t)} = x^{(t)}$, generate a proposal for $X^{(t+1)}$, i.e sample $Y^{(t)} \sim q(y \mid x^{(t)}$. Let us note $y^{(t)}$ the obtained value.
   - Compute the ratio

   $$\alpha(x^{(t)}, y^{(t)}) = \min\left\{1, \frac{\pi(y^{(t)})q(x^{(t)} \mid y^{(t)})}{\pi(x^{(t)}q(y^{(t)} \mid x^{(t)})}\right\}$$

   - Generate $U^{(t)} \sim U[0,1]$. Let us note $u^{(t)}$ the obtained value. Hence,

   $$X^{(t+1)} = \begin{cases} y^{(t)}, & \text{if } u^{(t)} < \alpha(x^{(t)}, y^{(t)}) \\ x^{(t)} & \text{otherwise} \end{cases}$$

3. $t \leftarrow t + 1$ and return to step 2.
   Point 2 is equivalent to accepting $X^{(t+1)} = y^{(t)}$ with probability $\alpha(x^{(t)}, y^{(t)})$, otherwise $X^{(t+1)} = x^{(t)}$.

Remark 2.3.1

- The sampling scheme shows that at each iteration, a sample $y^{(t)}$ is simulated according to the previous sample. The variables $\left(y^{(t)}\right)$, thus produced by the MH algorithm are not independent.
- The MH algorithm is specified by the proposal distribution $q(\cdot)$ whore choice is important.
- The simulation scheme is more efficient if $q\left(y \mid x^{(t)}\right)$ is quickly simulatable and close to π
- The target distribution $\pi(\cdot)$ is only used to compute the acceptance probability ratio $p = \dfrac{\pi(y(t))q(x(t) \mid y(t))}{\pi(x(t))} q\left(y^{(t)}x^{(t))}\right.$, and appears in both the numerator and the denominator.
- $\Rightarrow \pi$ can then just be known up to a multiplicative constant, which is convenient for bayesian estimation problems
- -In the MH algorithm, as in all MCMC methods, the samples from the first iterations of the algorithm are generally not used in estimating the functions
- They correspond to a period called burn-in, the number of iterations to come reasonably close to stationary.

Example 2.3.1

Let's consider as target distribution π a standard Cauchy distribution with a density

$$f_\pi(x) = \frac{1}{\pi(1+x^2)},\ x \in \mathbb{R}.$$

Let us assumes that it is known up to multiplicative constant:

$$f(x) \propto \frac{1}{1+x^2}.$$

We consider as a proposal a Normal whose mean is the previous value of the chain, and of fixed standard deviation $\sigma = 2$. We then have

$$q(y \mid x^{(t)}) = f_{\mathcal{N}}(y \mid x^{(t-1)}, \sigma).$$

Moreover, we have

$$q(x^{(t)} \mid y) = f_{\mathcal{N}}(x^{(t)} \mid y, \sigma),$$

where $f_{\mathcal{N}}$ is the density of the normal distribution, which can be written without the constants as

$$f_{\mathcal{N}}(x \mid \mu, \sigma) \propto \frac{1}{\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

Pratice: Matlab ex1 file.

- The Metropolis-Hastings algorithm. allows us to simulate samples according to a target distribution π from a candidate distribution 9 under fairly weals assumptions.
- Roughly speaking, any candidate distribution 9 in connected support can simulate any distribution n this support.
- Indeed, the simulation algorithms will not work well, for example, the high probability areas of the q density are mostly located in the tail of the target π.

2.3.3 The independent MH algorithm

The independent MH algorithm

- It is possible to generate a candidate variable y by going the present state of the chain $X^{(t)}$
- That is the proposal distribution is given by $q(y(x) = q(y)$
- We do get special ease of the original algorithm
- This algorithm can be written as:
  Given $x^{(t)}$

  1. Generate $Y^{(t)} \sim q(y)$.

2. Compute the ratio

$$\alpha(x^{(t)}, y^{(t)}) = \min\left\{1, \frac{\pi(y^{(t)})q(x^{(t)})}{\pi(x^{(t)})q(y^{(t)})}\right\}$$

3. Sample $U^{(t)} \sim U[0,1]$. The update now is

$$X^{(t+1)} = \begin{cases} y^{(t)} & \text{if } u^{(t)} < \alpha(x^{(t)}, y^{(t)}) \\ x^{(t)} & \text{otherwise} \end{cases}$$

- This method appears as a generalization of the **Accept-Reject** method in the sense that a is independent of $x^{(t)}$.
- Although the candidate variable $y^{(t)}$ is generated independently of $x^{(t)}$, the algorithm stick generates a sequence of correlated samples because the acceptance probability ratio depends on $x^{(t)}$ (Recall: the Accept - Reject samples are i.i.d)
- The MH will involve repeated occurrences of the same value since the rejection of $y^{(t)}$ leads to repetition of $x^{(t)}$ at time $(t+1)$
- The Accept-Reject method acceptance step requires the calculation of the upper bound $M \geqslant \sup_x \pi(x) \mid q(x)$, which is not required by the Metropolis-Hastings algorithm.
- This is an appealing feature of the Metropolis-Hastings if computing M is time-consuming, or if the existing M is inaccurate.
- Theoretically, it is feasible to use a pair (π, q) such that a bound M on $\pi/9$ does not exist, and thus to use Metropolis-Hastings, when Accept - Reject, is not possible.
- However, as illustrated in the following formal example, the performance of the MH algorithm can be very poor.

Example 2.3.2

- The forget distribution is the same as is the previous example, a standard Cauchy distribution with the density known up to a multiplicative constant $\gamma_\pi(x) \propto \dfrac{1}{1+x^2}$.
- It is possible, in theory, to consider as a proposal a standard normal $\mathcal{N}(0,1)$.
- We have $q\left(y \mid x^{(t)}\right) = f_{\mathcal{N}}(y \mid 0,1)$ and $q\left(x^{(t)} \mid y\right) = f_{\mathcal{N}}\left(x^{(t)} \mid 0,1\right)$ where $f_{\mathcal{N}}$ is the density (without the constant) given by:

$$\delta_{op}(x \mid 0,1) \propto \exp\left(\frac{-x^2}{2}\right)$$

Practice: Matlab ex2 file.

- Consider, for a second time, the initialization of the chain to a mere central value, for example, 0,5

- The following figure shows the density histogram overlaid with the target distribution and the trajectory of the chain.

The figure of the code here...

Example 2.3.3: Matlab ex3

The target distribution is still the same, but we now use a Students distribution with 0.5 dot.

- Even initialized to an extreme value ($1Q$), the behavior of the chain is completely different.
- Very large values of $x^{(t)}$ may occur from time to time as shown in the figures, but the histogram fit remains quite good.
- Mengersen and Tweedie (1996) established that good density families satisfy a dominance condition, as in the Accept-Reject algorithm, namely the existence of a constant M such that.

$$\pi(x) \leqslant M_q(x),$$

for all x in the support of π.

- This relation implies that proposal q must have a heavier tail than the target π, which allows to sweep of the whole support of π.
- The choice of proposal q in an MH algorithm is a complex problem, the subject of several discussions in the literature [Robber (1996), Gill (1996)].
- Typically, q is chosen from a family of distributions that allows one to specify the tuning parameters (location and scaling parameters) in such a way as to ensure the proper sweep of the target distribution support (as in the case of ex 1)
- We will again illustrate the importance of setting the parameters of the candidate law in the framework of the Metropolis-Hastings random walk algorithm.

2.3.4 Metropolis-Hastings random walk algorithm (symmetric MH)

Metropolis-Hastings random walk algorithm (symmetric MH)

- This is also a special case of the MH algorithms that consider a local exploration of the neighborhood of the current value of the Marker Chair.
- The candidate variables $y^{(t)}$ are generated in a neighborhood of the current value $x^{(t)}$ according to the relation.

$$y(t) = x^{(t)} + \epsilon_t$$

where ε_t is a random perturbation, $\varepsilon_t \perp x^{(t)}$, with distribution q (for instance a uniform on a normal).

- Since the candidate value is equal to the current value to which we add some perturbation, this algorithm is called random walk

- The proposal density is of the form q :

$$q(y \mid x)=q(y-x)=q(\varepsilon)$$

- If q is symmetric around $O($ i.e $q(z)=q(-z)$), the Markov chain associated with q is a random calla.
- But, due to the additional Metropolis-Hastings acceptance step, the Metropolis-Hastings Markov chain $X^{(t)}$ is not a random walk.
- In this symmetric case, the acceptance probability ratio becomes

$$\alpha\left(x^{(t)}, y^{(t)}\right)=\min \left\{1, \frac{\pi\left(y^{(t)}\right)}{\pi\left(x^{(t)}\right)}\right\}$$

and we get the original Metropolis' algorithm.

Remark 2.3.2

- In that case, the acceptance probability is driven by the objective ratio $\pi(y^{(t)})/\pi(x^{(t)})$ and is independent from q.
- This means that for a given pair $\left(x^{(t)}, y^{(t)}\right)$, the probability of acceptance is the same whether $y^{(t)}$, is generated from a Normal or from a Cauchy distribution.
- Obviously, changing q will result in different ranges of values for the y_t 's and a different acceptance rate, so this is not to say that the choice of q has no in pact whatsoever on the behavior of the algorithm.
- But this invariance of the acceptance probability is worth noting.
- In this algorithm, if q is positive in a neighborhood of 0, the chain is ergodic.
- This means that a prior, for any candidate verifying this condition, the algorithm will converge and generate samples according to π.
- But we are going to see that speed of convergence depends crucially on the parameters of the proposal distribution

Example 2.3.4: Matlab Ex4

To illustrate this, we consider an example where the target π is $\mathcal{N}(0,1)$ and take three proposals

1. $q(\cdot \mid x)=\mathcal{N}(\cdot \mid x, 0.5)$.
2. $q(\cdot \mid x)=\mathcal{N}(\cdot \mid x, 0.1)$.
3. $q(\cdot \mid x)=\mathcal{N}(\cdot \mid x, 10)$.

Comment:

- This figure describes three samples produced by this method for 3 different variances and clearly shows the difference in the produced chains.

- Chain 1 (top) exhibits good mixing properties ("mixes well"): it moves smoothly in the state space. Although initialized to an extreme value (-10), the algorithm converges quickly. The target distribution support (outlined by the red bands in the figures) is swept in 100 iterations.
- On the opposite, chains 2 and 3 mix badly despite an initialization to the mode of the target (here 0).
- Note the distinct patterns for chains 2 and 3.
- In the case of chain 2, the Markov chain moves at each iteration but very slowly. In the case of 3, it remains constant over long periods of time.
- This simple example illustrates the importance of the choice of the scale parameter of the candidate distribution (here the variance) in exploring the support of the target.
  1. If the variance is too small, the step size is small: the proposal generates a value of $y^{(t)}$ close to the current value $x^{(t)}$. The acceptance rate is high but the chain is poorly mixed (the support of the target is poorly explored). This is the case for chain 2.
  2. If the variance is too high, the generated candidate $y^{(t)}$ is too far from the current value $x^{(t)}$. The acceptance rate is therefore often low and the chain is badly mixed. The figure shows several iterations where chain 3 does not move and copies the previous values.
- As noted in this formal example, calibrating the scale of the random walk is crucial to achieving a good approximation to the target distribution in a reasonable number of iterations.

2.4 Gibbs Sampler

2.4.1 Introduction

- The Metropolis-Hastings algorithm can present difficulties in adequately exploring the support of the target distribution.
- These difficulties are significant when the distribution is multidimensional, because the number of samples necessary to have sufficient coverage of the support is then very important.
- The Gibbs sampling algorithm is particularly adapted to the simulation of multidimensional distributions.
- To simulate according to the target, the Gibbs sampler exploits its conditional distributions if they exist.

Miltistage Gibbs sampler

To sum up, the Gibbs sampling algorithm is used to simulate $\pi(X)$ when

- X can be partitioned as follows:

$$X = (X_1, X_2, ...X_N)$$

- The full conditionals

$$\pi_i(X_i \mid X_{-i})$$

are easily simulated, with the notation $X_{-i} = (X_1, X_2, \ldots, X_{i-1}, X_{i+1}, \ldots X_N)$ and $\pi_i\left(X_i \mid X_{-i}\right)$ is the conditional density of X_i given X_{-i}.

Sampling scheme of the Gibbs sampler

1. Iteration $t = 0$, initialize $\boldsymbol{X}^{(0)} \sim \pi_0(\boldsymbol{x})$ with $\boldsymbol{x} = (x_1, \ldots, x_N)$.

2. Iteration $t = 0, 1, \ldots$, sample

$$\begin{aligned}
X_1^{(t+1)} &\sim \pi_1\left(x_1 \mid x_2^{(t)}, \ldots, x_N^{(t)}\right)\\
X_2^{(t+1)} &\sim \pi_2\left(x_2 \mid x_1^{(t+1)}, x_3^{(t)} \ldots, x_N^{(t)}\right)\\
&\vdots\\
X_i^{(t+1)} &\sim \pi_i\left(x_i \mid x_1^{(t+1)}, \ldots, x_{i-1}^{(t+1)}, x_{i+1}^{(t)} \ldots, x_N^{(t)}\right)\\
&\vdots\\
X_N^{(t+1)} &\sim \pi_N\left(x_N \mid x_1^{(t+1)}, \ldots, x_{N-1}^{(t+1)}\right)
\end{aligned}$$

3. $t \leftarrow t + 1$ and return to step 2 .

Remark 2.4.1

- The Gibbs sampler requires to know and to be able to sample according to the conditional distributions.
- Unlike the MH algorithm, the conditionals are the only densities used for simulation.
- Thus, even in a high-dimensional problem, all of the simulations may be univariate, which is usually an advantage.
- $\rightarrow$ In fact, the x_i components are not necessarily scalar (block sampling).
- Unlike the MH, here all simulated samples are accepted.
- The simulation scheme of the Gibbs algorithm is, by construction, multi-dimensional. This algorithm is therefore only applicable to models with at least two random variables.
- In some cases, it is necessary to consider and simulate artificial variables for the implementation [see later]

Remark 2.4.2:

- As we will see later, the Gibbs sampler is particularly well adapted to hierarchical Bayesian model.
- The Gibbs sampler cannot be applied if the parameter vector to be simulated is of variable size.
  Solution: reversible jump MCMC or BNP methods

Implementing the Gibbs sampler requires the full conditional distributions (**FCD)**, that is, the conditional distribution of one parameter conditional on all the others. distribution of one parameter conditional on all the others.
Suppose for example $X = (X_S; X_{-s})$ and we are interested in the FCD of x_S. - The FCD is then

$$f\left(X_s \mid X_{-s}\right)=\frac{f\left(x_s, x_{-s}\right)}{\int f\left(x_s, X_{-s}\right) d x_s} \propto f\left(x_s, x_{-s}\right).$$

The FCD are conceptually easy to find.
We merely need to pick out all of the terms in the joint distribution that involve that parameter.
Two - stage Gibbs sampler

- To fix the ideas, we consider the 2D case where $X = (X_1, X_2) := (X, Y)$, with joint density $f(x, y)$.
- Starting from $X^{(0)} = x^{(0)}$ and $Y^{(0)} = y^{(0)}$, the principle of the two-stage Gibbs sampler is, successively, to simulate $X^{(1)} \mid Y^{(0)} = y^{(0)}$ and $Y^{(1)} \mid X^{(1)} = x^{(1)}$, then $X^{(2)}Y$ distributions.
- This Algorithm is then straightforward to implement as long as simulating from both conditionals is feasible.
- Convergence of Markov chain (and thus the algorithm) is ensured unless the supports of the conditionals are not connected.
- The sequence $(X^{(t)}, Y^{(t)})_t$ produced by the Gibbs sampler converges to the joint distribution f and, as a consequence, the sequences $(X^{(t)})_t$ (respectively $(Y^{(t)})_t$ tends to the marginal distribution of X (respectively Y) when $t \to \infty$.
- Using $\mathbb{E}[f(x \mid Y)] = \displaystyle\int f(x \mid y) f(y) dy = f(X)$
- An estimate of the marginal $f(x)$ can be obtained by:

  $$d(x)=\frac{1}{n-m} \sum_{t=m+1}^n f\left(x \mid y^{(t)}\right) \quad (1)$$

  where $y^{(m+1)}, y^{(m+2)}, \ldots . y^{(n)}$ is a final sequence of the Gibbs sampler ($m =$ burn-in period and n total number of iterations), and thus approximates a sample of $f(y)$.

Example 2.4.1

Let's consider the following joint distribution of X and Y :

$$f(x,y) \propto \begin{pmatrix} n \\ x \end{pmatrix} y^{x+\alpha-1}(1-y)^{n-x+\beta-1}$$

for $x = 0, 1, \ldots,$ n and $0 \leq y \leq 1$.

1. Calculate the FCD of X and Y.
2. The Matlab code **Example1 Gibbs 2D beta binomial.m** implements the sampling scheme, by considering $n = 16, \alpha = 2$ and $\beta = 4$.

Solution:

1. We have

- $f(x,y) = C_n^x y^{x+\alpha-1}(1-y)^{n-x+\beta-1}$
- $f(x|y) = C_n^x y^x (1-y)^{n-x} \Rightarrow X|y \sim B(n,y)$
- $f(y|x) = y^{x+\alpha-1}(1-y)^{n-x+\beta-1} \Rightarrow Y|x \sim \text{Beta}(x+\alpha, n-x+\beta)$

Remark 2.4.3

- This is a toy example because the Gibbs sampler is not needed to estimate $p(x)$.
- Indeed, the latter has a closed-form and is given by:

$$p(x) = \frac{f(x,y)}{f(y \mid x)} = \begin{pmatrix} n \\ x \end{pmatrix} \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \frac{\Gamma(x+\alpha)\Gamma(n-x+\beta)}{\Gamma(\alpha+\beta+n)},$$

- This is a less-standard distribution known as Beta-binomial distribution.
- We can compute the estimated marginal of $p(x)$, and compare it to this true marginal.

Example 2.4.2

Let's now consider the bivariate Normal model: $(X, Y) \sim \mathcal{N}(\mu, \Sigma)$,

- The FCD $X \mid Y = y$ and $Y \mid X = X$ are also Gaussian and can be written as:

$$\begin{aligned}
(X \mid Y = y) &\sim \mathcal{N}\left(\rho Y, 1-\rho^2\right) \\
(Y \mid X = X) &\sim \mathcal{N}\left(\rho X, 1-\rho^2\right)
\end{aligned}$$

- The sub chain $X^{(1)}$ satisfies $X^{(1+1)} \mid X^{(1)} = X_1 \sim \mathcal{N}\left(\rho^2 X_1, 1-\rho^4\right)$ and a recursion shows that $X^{(1)} \mid X^{(0)} = x_0 \sim \mathcal{N}\left(\rho^{21} X_0, 1-\rho^{4t}\right)$, which does indeed converge to a $\mathcal{N}(0,1)$ as t goes to ∞.
- The principle of the Gibbs sampler is, starting from $\left(x^{(0)}, y^{(0)}\right)$, generate successively $X^{(1)} \mid y^{(0)}$ and $Y^{(1)} \mid x^{(1)}$, then $X^{(2)} \mid y^{(1)}$ and $Y(2) \mid X^{(2)}$, and so using the conditionals.
- Ct Example2 Gibbs bivariate normal.m.

Example 2.4.3

- We start with a simple case of a Gaussian sample with unknown mean μ and unknown precision τ.
- The prior tor μ is astandard Normal distribution, and the prior for a a Gamma with parameters 2 and 1. The hierarchical model is written as follows:

$$\begin{aligned}
&x_1, x_2, \ldots, x_n \sim \mathcal{N}(\mu, 1/\tau) \\
&\mu \sim \mathcal{N}(0,1) \\
&\tau \sim G(2,1).
\end{aligned}$$

1. Compute the joint distribution $f(x, \mu, \tau)$.
2. Compute the FCD of μ and the FCD τ.

Solution:

1. The joint distribution is computed as follows

$$\begin{aligned}
f(x, \mu, \tau) &= f(x|\mu, \tau)\pi(\mu)\pi(\tau) \\
&= \pi(\mu)\pi(\tau)\prod_{i=1}^{n} f(x_i|\mu, \tau) \\
&\propto \exp\left\{-\frac{\mu^2}{2}\right\} \times \tau^{2-1}e^{-\tau} \times \prod_{i=1}^{n} \sqrt{\tau}\exp\left\{-\frac{(x_i-\mu)^2}{2\dfrac{1}{\tau}}\right\} \\
&\propto \tau^{n/2+1}\exp\left\{\frac{-\mu^2 - 2\tau - \tau\sum_{i=1}^{n}(x_i-\mu)^2}{2}\right\}
\end{aligned}$$

2. The conditional of τ is

$$\begin{aligned} f(\tau|x,\mu) &\propto f(\tau,x,\mu) \\ &\propto \tau^{n/2+1}\exp\left\{-\frac{\mu^2+2\tau+\tau\sum_{i=1}^n(x_i-\mu)^2}{2}\right\} \\ &\propto \tau^{n/2+1}\exp\left\{-\tau\left(1+\frac{1}{2}\sum_{i=1}^n(x_i-\mu)^2\right)\right\} \end{aligned}$$

Then $\tau|x,\mu \sim \text{Gamma}\left(\frac{n}{2}+2, 1+\frac{1}{2}\sum_{i=1}^n(x_i-\mu)^2\right)$.
On the other hand, the conditional probability of μ is computed as follows

$$\begin{aligned} f(\mu|x,\tau) &\propto f(\mu,x,\tau) \\ &\propto \exp\left\{-\frac{\mu^2+n\tau\mu^2-2\mu\tau\sum_{i=1}^n x_i}{2}\right\} \\ &\propto \exp\left\{-\frac{\mu^2-2\mu\frac{\tau}{n\tau+1}\sum_{i=1}^n x_i}{2\frac{1}{n\tau+1}}\right\} \\ &\propto \exp\left\{-\frac{\left(\mu-\frac{\tau}{n\tau+1}\sum_{i=1}^n x_i\right)^2}{2\frac{1}{n\tau+1}}\right\} \end{aligned}$$

Therefore, $\mu|x,\tau \sim \mathcal{N}\left(\frac{\tau}{n\tau+1}\sum_{i=1}^n x_i, \frac{1}{n\tau+1}\right)$.

2.4.2 Mixture model

- Mixture models are widely used in Statistics, pattern recognition and data mining.
- We focus on the Gaussian mixture model.
- Assume that a sample $X_1,\ldots,X_n$ comes from the following mixture:

$$f\left(x \mid w,\mu,\sigma^2\right)=\sum_{j=1}^K w_j f_{\mathcal{N}}\left(x \mid \mu_j,\sigma^2\right)$$

  - $W=(W_1,\ldots,w_K)$ is the vector containing the unknown weights. These weights verity $0\le w_j\le 1$ and $\sum_{l=1}^K w_j=1$.
  - The vector containing the unknown means is $\mu=(\mu_1,\ldots,\mu_k)$, and $\sigma^2>0$ is is the unknown common variance.
- The **estimation problem** is to determine the mean, the amplitude (weight) of each Gaussian, and their common variance.
- These parameters can be estimated using the EM algorithm to maximize the likelihood.

- We consider here the Bayesian treatment.
- We endow the unkown parameters with priors according to the following model:

$$
\begin{aligned}
& w \sim \operatorname{Dir}(\alpha_1, \ldots, \alpha_K) \\
& \mu_i \sim \mathcal{N}\left(0, \tau^2\right), j = 1, \ldots, K \\
& \sigma^2 \sim \mathcal{IG}(\gamma, \delta).
\end{aligned}
$$

- The Dirichlet distribution is a generalization of the Beta.
- Let $Y = (y_1, \ldots, y_K)$ and $\alpha = (\alpha_1, \ldots, \alpha_K)$ with $\forall i, \alpha_i > 0$, $y_i > 0$ and $\sum_{i=1}^{K} y_i = 1$. If $Y \sim \operatorname{Dir}(\alpha_1, \ldots, \alpha_K)$, its p.d.f. is

$$
f(y_1, \ldots, y_k \mid \alpha_1, \ldots, \alpha_k) = \frac{1}{B(\alpha)} \prod_{i=1}^{K} y_i^{a_j - 1}
$$

where $B(\alpha)$ is the multinomial beta function:

$$
B(\alpha) = \frac{\prod_{i=1}^{K} \Gamma(a_i)}{\Gamma\left(\sum_{i=1}^{K} a_i\right)}
$$

2.4.3 Bayesian Mixture Model

We consider here that parametrization of the Dirichelet

$$
f(y_1, \ldots, y_k \mid \alpha_1, \ldots, \alpha_K) \propto \prod_{i=1}^{k} y_i^{\alpha_i}
$$

The joint distribution is

$$
\begin{aligned}
f\left(x, w, \mu, \sigma^2\right) &= f\left(x \mid w, \mu, \sigma^2\right) \pi(\mu) \pi(w) \pi\left(\sigma^2\right) \\
&\propto \left\{ \prod_{i=1}^{n} \sum_{j=1}^{k} w_j \frac{1}{\sigma} \exp\left(-\frac{1}{2\sigma^2} (x_i - \mu)^2\right) \right\} \\
&\left\{ \prod_{j=1}^{K} \exp\left(-\frac{\mu_j^2}{2\tau^2}\right) \right\} \left\{ \prod_{j=1}^{k} w_j^a \right\} \\
&\left\{ \left(\frac{1}{\sigma^2}\right)^{\gamma+1} \exp\left(\frac{-\delta}{\sigma^2}\right) \right\}
\end{aligned}
$$

- Rather than dealing with this complex distribution, we introduce, as in the EM algorithm, the so-called **classification** or **indicator variables**.
- Let $c_i \in \{1, ..., K\}$ be the variable indicating the distribution from which the observation i is coming: $c_i = j$ if observation x_i comes from the distribution j.
- We say that we **augment the data**$(x_1, ..., x_n)$ by an unobservable n vector $c = (c_1, ..., c_n$.

- We can therefore replace the likelihood

$$f\left(x_i \mid w, \mu, \sigma^2\right)=\sum_{j=1}^k w_j \mathcal{N}\left(x_i \mid \mu_j, \sigma^2\right),$$

with

$$\left\{\begin{array}{l}X_i \mid c_i=j, \mu_j, \sigma^2 \sim N\left(x_i \mid \mu j, \sigma_j^2\right) \\ \mathbb{P}(c_i=j \mid w)=w_j \leftrightarrow \pi\left(c_i \mid w\right)=w_{c_i}.\end{array}\right.$$

- Thus, considering (x_i, c_i) (instead of x_i) entirely eliminates the mixture structure. We can re-write

$$\pi(c_i \mid w)=w_1^{\delta_{c_{i,1}}} w_2^{\delta_{c_{i,2}}} \ldots w_K^{\delta_{c_{i,K}}}=\prod_{j=1}^K w_j^{\delta_{c_{i,j}}}$$

where $\delta_{c_{i,j}}=[c_i=j]$ is the Kronecker delta.

- Since the c_i are independent and identically distributed, we can write:

$$\pi(\boldsymbol{c} \mid \boldsymbol{w})=\prod_{i=1}^n \pi\left(c_i \mid \boldsymbol{w}\right)=\prod_{i=1}^n \prod_{j=1}^K w_j^{\delta_{c_{i,j}}}=\prod_{j=1}^K w_j^{\sum_{i=1}^n \delta_{c_{i,j}}}=\prod_{j=1}^K w_j^{n_j}$$

where $n_j=\#\left\{c_i: c_i=j, i=1, \ldots, n\right\}$ denotes the number of observations coming from the j-th density (we will also say the j-th component).

- Considering this augmented model, the joint distribution is given by:

$$\begin{aligned} f\left(\boldsymbol{x}, \boldsymbol{c}, \boldsymbol{w}, \boldsymbol{\mu}, \sigma^2\right) &= f\left(\boldsymbol{x} \mid \boldsymbol{c}, \boldsymbol{w}, \boldsymbol{\mu}, \sigma^2\right) \pi(\boldsymbol{c} \mid \boldsymbol{w}) \pi(\boldsymbol{w}) \pi(\boldsymbol{\mu}) \pi\left(\sigma^2\right) \\ &\propto \frac{1}{\sigma^n} \exp \left\{\frac{-1}{2 \sigma^2} \sum_{i=1}^n\left(x_i-\mu_{c_i}^2\right)\right\} \times \prod_{i=1}^n w_{c_i} \times \prod_{j=1}^K w_j^\alpha \\ &\times \exp \left\{\frac{-1}{2 \tau^2} \sum_{j=1}^K \mu_j^2\right\} \times \frac{1}{\sigma^{2(\gamma+1)}} \exp \left\{\frac{-\delta}{\sigma^2}\right\} \end{aligned}$$

with $\prod_{j=1}^K w_j^{n_j}=\prod_{i=1}^n w_{c_i}$.

- To apply the Gibbs sampler we need to find the full conditionals.

- **Compute the FCD of w, μ, σ^2, and c.**

- FCD of w.
By choosing in the joint distribution the terms containing $\boldsymbol{w}$, we have

$$\pi\left(\boldsymbol{w} \mid \boldsymbol{\mu}, \sigma^2, \boldsymbol{c}, \boldsymbol{x}\right) \propto \prod_{i=1}^n w_{c_i} \prod_{j=1}^K w_j^\alpha=\prod_{j=1}^K w_j^{n_j} \prod_{j=1}^K w_j^\alpha=\prod_{j=1}^K w_j^{\alpha+n_j}.$$

Thus, $\boldsymbol{w} \mid \boldsymbol{\mu}, \sigma^2, \boldsymbol{c}, \boldsymbol{x} \sim \operatorname{Dir}\left(\alpha+n_1, \ldots, \alpha+n_K\right)$.

- FCD of σ^2.
We have $\pi\left(\sigma^2 \mid \boldsymbol{w}, \mu, \boldsymbol{c}, \boldsymbol{x}\right) \propto$

$$
\begin{aligned}
&\frac{1}{\sigma^n} \exp\left\{\frac{-1}{2\sigma^2}\sum_{i=1}^{n}\left(x_i-\mu_{c_i}\right)^2 \times\right\} \frac{1}{\sigma^{2(\gamma+1)}} \exp\left\{\frac{-\delta}{\sigma^2}\right\}\\
&=\frac{1}{\left(\sigma^2\right)^{n/2+\gamma+1}} \exp\left\{\frac{-1}{\sigma^2}\left(\delta+\frac{\sum_{i=1}^{n}\left(x_i-\mu_{c_i}\right)^2}{2}\right)\right\}\\
&\propto \mathcal{IG}(\alpha,\beta)
\end{aligned}
$$

with $\alpha=\gamma+\dfrac{n}{2}$ and $\beta=\delta+\dfrac{1}{2}\displaystyle\sum_{i=1}^{n}\left(x_i-\mu_{c_i}\right)^2$.

- FCD of $\boldsymbol{\mu}$.
One has:

$$
\begin{aligned}
&\pi\left(\mu \mid \boldsymbol{w}, \sigma^2, \boldsymbol{c}, \boldsymbol{x}\right) \propto \exp\left\{\frac{-1}{2\sigma^2}\sum_{i=1}^{n}\left(x_i-\mu_{c_i}^2\right)\right\}\exp\left\{\frac{-1}{2\tau^2}\sum_{j=1}^{K}\mu_j^2\right\}\\
&=\exp\left\{\frac{-1}{2\sigma^2}\sum_{j=1}^{K}\sum_{i:c_i=j}\left(x_i-\mu_j^2\right)\right\}\exp\left\{\frac{-1}{2\tau^2}\sum_{j=1}^{K}\mu_j^2\right\}\\
&\propto \exp\left\{\frac{-1}{2\sigma^2}\sum_{j=1}^{K}\left(-2\mu_j\sum_{i:c_i=j}x_i+n_j\mu_j^2\right)-\frac{1}{2\tau^2}\sum_{j=1}^{K}\mu_j^2\right\}
\end{aligned}
$$

- The FCD of μ_j.

$$
\begin{aligned}
&\pi\left(\boldsymbol{\mu} \mid \boldsymbol{w}, \sigma^2, \boldsymbol{c}, \boldsymbol{x}\right) \propto \exp\left\{\sum_{j=1}^{K}\left[\mu_j^2\left(-\frac{n_j}{2\sigma^2}-\frac{1}{2\tau^2}\right)+\frac{\mu_j}{\sigma^2}\sum_{i:c_i=j}x_i\right]\right\}\\
&\propto \prod_{j=1}^{K}\exp\left\{\frac{-1}{2}\left(\frac{n_j}{\sigma^2}+\frac{1}{\tau^2}\right)\left[\mu_j^2-\frac{2\mu_j\left(\frac{1}{\sigma^2}\sum_{i:c_i=j}x_i\right)}{\frac{n_j}{\sigma^2}+\frac{1}{\tau^2}}\right]\right\}\\
&\propto \prod_{j=1}^{K}\exp\left\{\frac{-1}{2}\left(\frac{n_j}{\sigma^2}+\frac{1}{\tau^2}\right)\left[\mu_j-\frac{1/\sigma^2\sum_{i:c_i=j}x_i}{\frac{n_j}{\sigma^2}+\frac{1}{\tau^2}}\right]^2\right\}\\
&\propto \prod_{j=1}^{K}\mathcal{N}\left(\mu_j \mid m_j, v_j\right)
\end{aligned}
$$

- The full conditional on μ is multivariate normal, with independent components μ_j.

- Thus $\mu_j \mid \boldsymbol{w}, \sigma^2, \boldsymbol{c}, \boldsymbol{x} \overset{iid}{\sim} \mathcal{N}\left(\mu_j \mid m_j, v_j\right), j=1,\ldots,K$ where $v_j=\left(\dfrac{n_j}{\sigma^2}+\dfrac{1}{\tau^2}\right)^{-1}$ and $m_j=\dfrac{\tau^2}{n_j\tau^2+\sigma^2}\displaystyle\sum_{i:c_i=j}x_i$.

- FCD of $\boldsymbol{c}$:

$$\begin{aligned}\pi\left(\boldsymbol{c} \mid \boldsymbol{w}, \mu, \sigma^2 \boldsymbol{x}\right) &\propto \exp\left\{\frac{-1}{2\sigma^2}\sum_{i=1}^{n}\left(x_i-\mu_{c_i}\right)^2\right\} \times \prod_{i=1}^{n} w_{c_i} \\ &= \prod_{i=1}^{n} w_{c_i} \exp\left(\frac{-1}{2\sigma^2}\left(x_i-\mu_{c_i}\right)^2\right)\end{aligned}$$

By independence of the c_i, we thus have $\forall i = 1, \ldots, n$,

$$\mathbb{P}\left(c_i = j\right) \propto w_j \exp\left(\frac{-1}{2\sigma^2}\left(x_i-\mu_j\right)^2\right)$$

2.4.4 Change-point model

- An extension between mixture models and hidden Markov chains is change-point models, where the parameter of the distribution of a temporal (or causal) phenomenon changes at one (or more) unknown time(s).

- A change-point detection model tries to identify the time(s) during which the change occurs.

- The simplest version of a change-point model is the observation of a time series $(x_1, \ldots, x_n)$, associated with a latent index τ, such that:

$$\tau \sim \pi_0(\tau), \quad x_1, \ldots, x_\tau \Big| \tau \overset{iid}{\sim} f\left(x \mid \xi_1\right), \quad x_{\tau+1}, \ldots, x_n \Big| \tau \overset{iid}{\sim} f\left(x \mid \xi_2\right),$$

and π_0 has support $\{1, \ldots, n\}$.

- The observations can therefore be divided into two subsets $(x_1, \ldots, x_\tau)$ and $(x_{\tau+1}, \ldots, x_n)$.

Example 2.4.4: Poissonian Data (the change time is known)

- We consider piecewise constant Poissonian data of the form:

$$\begin{cases} X_i \mid \lambda_1 & \overset{\text{i.i.d}}{\sim} \mathcal{P}(\lambda_1) \quad \text{for } i = 1, \ldots, l_1 \\ X_i \mid \lambda_2 & \overset{\text{i.i.d}}{\sim} \mathcal{P}(\lambda_2) \quad \text{for } i = (l_1+1), \ldots, n. \end{cases}$$

  We assume l_1 is known, in this first example.

- We endow the model parameters with the following priors

$$\begin{cases} \lambda_1 \mid \beta & \sim \mathcal{G}(\alpha, \beta), \\ \lambda_2 \mid \beta & \sim \mathcal{G}(\alpha, \beta), \end{cases}$$

  with $\alpha = 2$ but β is unknown.

- For the last level of the hierarchy, we consider the non-informative prior on the hyperparameter β :

$$\pi(\beta) = \frac{1}{\beta} \mathbf{1}_{\mathbb{R}^+}(\beta).$$

- To evaluate this model, we consider $n = 200$ observations generated according to a Poisson distribution with parameters $\lambda_1 = 20$ and $\lambda_2 = 13$, with $l_1 = 100$.

- See file Poissonian data and function simu-poisson.m We consider 5, 000 Monte-Carlo iterations and discard the first 250 for the burn-in.

- The figure shows the theoretical conditionals overlaid with the estimated ones (via the Gibbs sampler) respectively for λ_1, λ_2 and β

- Note: to compute the theoretical conditionals of λ_1, λ_2, we use the true values of λ_1, λ_2, and plug the estimated value of β.

- **Compare the estimated values of λ_1, λ_2 with the true ones**.

Solution:

- Compute the joint distribution $f(\mathbf{x}, \lambda_1, \lambda_2, \beta)$, with $\boldsymbol{x} = \left(\boldsymbol{x}^{(1)}, \boldsymbol{x}^{(2)}\right)$ where $\boldsymbol{x}^{(1)} = (x_1, \ldots, x_{l_1})$ and $\boldsymbol{x}^{(2)} = (x_{1+1}, \ldots, x_n)$. The joint distribution is

$$\begin{aligned} f(\boldsymbol{x}, \lambda_1, \lambda_2, \beta) &= f\left(\boldsymbol{x}^{(1)} \mid \lambda_1\right) f\left(\boldsymbol{x}^{(2)} \mid \lambda_2\right) \pi(\lambda_1 \mid \beta)\, \pi(\lambda_2 \mid \beta)\, \pi(\beta) \\ &\propto \left[\prod_{i=1}^{l_1} e^{-\lambda_1} \frac{\lambda_1^{x_i}}{x_i!}\right] \left[\prod_{i=l_1+1}^{n} e^{-\lambda_2} \frac{\lambda_2^{x_i}}{x_i!}\right] \left[\beta^\alpha \lambda_1^{\alpha-1} e^{-\beta\lambda_1}\right] \left[\beta^\alpha \lambda_2^{\alpha-1} e^{-\beta\lambda_2}\right] \frac{1}{\beta} \mathbf{1}_{\mathbb{R}^+}(\beta) \\ &= e^{-\lambda_1 l_1} \frac{\lambda_1^{\sum_{i=1}^{l_1} x_i}}{\prod_{i=1}^{l_1} x_i!} e^{-\lambda_2(n-l_1)} \frac{\lambda_2^{\sum_{i=1_1+1}^{n} x_i}}{\prod_{i=l_1+1}^{n} x_i!} \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda_1^{\alpha-1} e^{-\beta\lambda_1} \times \beta^\alpha \lambda_2^{\alpha-1} e^{-\beta\lambda_2} \frac{1}{\beta} \mathbf{1}_{\mathbb{R}^+}(\beta). \end{aligned}$$

- Deduce the FCD of λ_1, λ_2 and β.
  FCD of λ_1.

$$\pi(\lambda_1 \mid \beta, x) \propto e^{-\lambda_1 l_1} \lambda_1^{\sum_{i=1}^{l_1} x_i} \lambda_1^{\alpha-1} e^{-\beta\lambda_1} = \lambda_1^{\sum_{i=1}^{l_1} x_i + \alpha - 1} e^{-(l_1+\beta)\lambda_1}$$

Hence, $\lambda_1 \mid \beta, x \sim \text{Gamma}\left(\sum_{i=1}^{l_1} x_i + \alpha, \beta + l_1\right)$.

FCD of λ_2 is

$$\pi(\lambda_2 \mid \beta, x) \propto e^{-\lambda_2(n-l_1)}\lambda_2^{\sum_{i=l_1+1}^{n} x_i} \times \lambda_2^{\alpha-1}e^{-\beta\lambda_2} = \lambda_2^{\sum_{i=l_1+1}^{n} x_i+\alpha-1}e^{-(n-l_1+\beta)\lambda_2}$$

Hence $\lambda_2 \mid \beta, x \sim \text{Gamma}\left(\sum_{i=l_1+1}^{n} x_i + \alpha, n - l_1 + \beta\right)$.

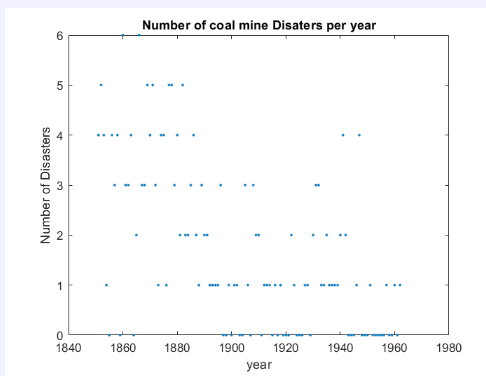
FCD of β is

$$\pi(\beta \mid \lambda_1, \lambda_2, x) \propto \beta^{\alpha}e^{-\beta\lambda_1}\beta^{\alpha}e^{-\beta\lambda_2}\frac{1}{\beta}1_{\mathbb{R}^+}(\beta) = \beta^{2\alpha-1}e^{-(\lambda_1+\lambda_2)\beta}$$

Hence, $\beta \mid \lambda_1, \lambda_2, x \sim \text{Gamma}\,(2\alpha, \lambda_1 + \lambda_2)$

Example 2.4.5: Poissonian Data (the time change is unknown)

- We are now interested in longitudinal count data.
- In this case data in the form of a time series relating to the annual number of coal mine disasters in England from 1851 to 1962.
- A plot of these data is given below and the time series itself is stored in the file coal.mat



- There is a total of $n = 112$ observations.
- From this plot, it does seem to be the case that there has been a reduction in the rate of disasters over the period in question.
- We assume there has been an abrupt change in a year (corresponding to e.g. the introduction of new legislation on security measurements).
- We would like to answer the following questions:
  1. What is the most likely year in which the change occurred?
  2. Did the disaster rate increase or decrease after the change?

- The annual number of accidents can be modeled as a Poisson distribution with parameter θ until the k-th year.

- After that, the number of accidents is distributed according to a Poisson distribution with parameter λ.

- This can be represented by the following model:

$$\begin{cases} Y_i \sim \mathcal{P}(\theta) & \text{for } i = 1, \ldots, k \\ Y_i \sim \mathcal{P}(\lambda) & \text{for } i = k+1, \ldots, n \end{cases}$$

  where k is the **unknown** change time.

- The priors for the parameters are given as follows:

$$\begin{cases} \theta & \sim \mathcal{G}(a_1, b_1), \\ \lambda & \sim \mathcal{G}(a_2, b_2), \\ k & \sim \mathcal{U}\{1, \ldots, 112\}, \end{cases}$$

- Note: k is distributed according to the discrete uniform distribution on $\{1, \ldots, 112\}$ because there are 112 years.

- The parameters θ, λ, and k are independent between them.

- We consider the following priors for the hyperparameters b_1 and b_2:

$$\begin{cases} b_1 \sim \mathcal{G}(c_1, d_1) \\ b_2 \sim \mathcal{G}(c_2, d_2) \end{cases}$$

  where c_1, d_1, c_2, d_2 are fixed.

Solution:

- Compute the joint distribution $f(y, \theta, \lambda, k, b_1, b_2)$.
  The joint distribution is:

$$\begin{aligned} f(\boldsymbol{y}, \theta, \lambda, k, b_1, b_2) \propto & \left(\prod_{i=1}^{k} e^{-\theta} \theta^{y_i}\right) \left(\prod_{i=k+1}^{n} e^{-\lambda} \lambda^{y_i}\right) \left(b_1^{a_1} \theta^{a_1 - 1} e^{-b_1 \theta}\right) \left(b_2^{a_2} \lambda^{a_2 - 1} e^{-b_2 \lambda}\right) \\ & \times \mathbf{1}_{\{1, \ldots, 112\}}(k) b_1^{c_1 - 1} e^{-d_1 b_1} b_2^{c_2 - 1} e^{-d_2 b_2} \\ & = e^{-k\theta} \theta^{\sum_{i=1}^{k} y_i} e^{-(n-k)\lambda} \lambda^{\sum_{i=k+1}^{n} y_i} \times b_1^{a_1} \theta^{a_1 - 1} e^{-b_1 \theta} b_2^{a_2} \lambda^{a_2 - 1} e^{-b_2 \lambda} \\ & \times \mathbf{1}_{\{1, \ldots, 112\}}(k) b_1^{c_1 - 1} e^{-d_1 b_1} b_2^{c_2 - 1} e^{-d_2 b_2} \end{aligned}$$

- Compute the FCD of θ, λ, b_1, b_2 and k. (**Tricks:** introduce $\dfrac{\lambda^{\sum_{i=1}^{k} y_i}}{\lambda^{\sum_{i=1}^{k} y_i}}$ when computing the FCD of k.

To sum up, the conditionals are:

$$\begin{gathered}
\theta \mid \boldsymbol{y}, \lambda, b_1, b_2, k \sim \mathcal{G}\left(a_1 + \sum_{i=1}^{k} y_i, k + b_1\right) \\
\lambda \mid \boldsymbol{y}, \theta, b_1, b_2, k \sim \mathcal{G}\left(a_2 + \sum_{i=k+1}^{n} y_i, n - k + b_2\right) \\
b_1 \mid \boldsymbol{y}, \theta, \lambda, b_2, k \sim \mathcal{G}\left(a_1 + c_1, \theta + d_1\right) \\
b_2 \mid \boldsymbol{y}, \theta, \lambda, b_1, k \sim \mathcal{G}\left(a_2 + c_2, \lambda + d_2\right) \\
f\left(k \mid \boldsymbol{y}, \theta, \lambda, b_1, b_2\right) = \frac{\mathcal{L}(y \mid k, \theta, \lambda)}{\sum_{j=1}^{n} \mathcal{L}(y \mid j, \theta, \lambda)} \\
\text{where } \mathcal{L}(y \mid k, \theta, \lambda) = \exp\{k(\lambda - \theta)\}(\theta/\lambda)^{\sum_{i=1}^{k} y_i}
\end{gathered}$$

Details for computing FCD of k:

$$\begin{aligned}
\pi(k \mid \ldots) &\propto e^{-k\theta} \theta^{\sum_{i=1}^{k} y_i} e^{-(n-k)\lambda} \lambda^{\sum_{i=k+1}^{n} y_i} \mathbf{1}_{\{1,\ldots,112\}}(k) \\
&= e^{-k\theta - n\lambda + k\lambda} \theta^{\sum_{i=1}^{k} y_i} \lambda^{\sum_{i=k+1}^{n} y_i} \mathbf{1}_{\{1,\ldots,112\}}(k) \\
&\propto e^{k(\lambda-\theta)} \theta^{\sum_{i=1}^{k} y_i} \lambda^{\sum_{i=k+1}^{n} y_i} \mathbf{1}_{\{1,\ldots,112\}}(k) \left(e^{-n\lambda} \perp k\right) \\
&= e^{k(\lambda-\theta)} \frac{\lambda^{\sum_{i=1}^{k} y_i}}{\lambda^{\sum_{i=1}^{k} y_i}} \theta^{\sum_{i=1}^{k} y_i} \lambda^{\sum_{i=k+1}^{n} y_i} \mathbf{1}_{\{1,\ldots,112\}}(k) \\
&= e^{k(\lambda-\theta)} \left(\frac{\theta}{\lambda}\right)^{\sum_{i=1}^{k} y_i} \underbrace{\lambda^{\sum_{i=1}^{k} y_i} \lambda^{\sum_{i=k+1}^{n} y_i}}_{=\lambda^{\sum_{i=1}^{n} y_i}} \mathbf{1}_{\{1,\ldots,112\}}(k) \\
&= e^{k(\lambda-\theta)} \left(\frac{\theta}{\lambda}\right)^{\sum_{i=1}^{k} y_i} \lambda^{\sum_{i=1}^{n} y_i} \mathbf{1}_{\{1,\ldots,112\}}(k) \\
&\propto e^{k(\lambda-\theta)} \left(\frac{\theta}{\lambda}\right)^{\sum_{i=1}^{k} y_i} \mathbf{1}_{\{1,\ldots,112\}}(k) \quad \text{since } \lambda^{\sum_{i=1}^{n} y_i} \perp k.
\end{aligned}$$

- Answer the required questions

1. - From the figure of the distribution of the variables k generated by the sampler, we can see that the mode is at $k = 41$, which corresponds to the year 1891 .
   - To quantify, we can compute $\mathbb{P}(k = 41)$. $\hookrightarrow$ $\mathbb{P}(k = 41) \approx 0.248$ since 248 on the 1000 post-burn-in drawn values equal 41 .

2. -From the histograms of the posterior distributions of θ and λ, it seems clear that there was a reduction in the average rate of accidents per year after this period.
   - Also, we can compute $\mathbb{P}(\theta > \lambda \mid \boldsymbol{y})$. $\hookrightarrow$ This is roughly 1 since $\theta^{(t)}$ was greater than $\lambda^{(t)}$ for all of the post-burn in sampled pairs $\left(\theta^{(t)}, \lambda^{(t)}\right)$.

Remark 2.4.4

- Mixture models are an example of missing-data models.
- We completed the data by auxiliary (classification) variables to aid in the simulation.
- There are many settings where a natural completion does exist.
- **Censored-data model** can also be treated as a missing-data model.

Example 2.4.6: Censored data

- In some cases, due to measurement or experimental constraints, the ”full” data is unavailable, having been coarsened, truncated, or otherwise censored.
- A concrete example of **censored data** is survival analysis, which aims to determine the effect of a drug on lifespan.
- If the study proceeds for a fixed amount of time, surviving subjects represent truncated data in the sense that only a minimum lifespan is known when the study concludes.
- In such cases, it may be difficult to sample from the posterior for the parameter of interest directly.
- Augmenting with variables representing the true values of the truncated or coarsened data, coupled with Gibbs sampling, may provide posterior samples.

Example 2.4.7

- For 360 consecutive time units, consider recording the number of passages of individuals per unit time in front of some sensor
- For instance, this can be the number of cars passing a crossroad, or the number of leucocytes on a grid of a blood sample.
- The results are summarized below:

Number of passages	0	1	2	3	4 or more
Number of observations	139	128	55	25	13

- This table, therefore, performs a grouping of the observations with four passages or more.

- We assume that the ungrouped observations are $X_i \sim P(\lambda)$.
- The likelihood of the corresponding model is

$$\mathcal{L}_{\boldsymbol{X}}(\lambda) \propto e^{-347\lambda}\lambda^{313}\left(1 - e^{-\lambda}\sum_{i=0}^{3}\frac{\lambda^i}{i!}\right)^{13}$$

- We endow λ with a prior $\pi(\lambda) = \dfrac{1}{\lambda}$.

- We complete the model with the vector $\boldsymbol{y} = (y_1, \ldots, y_{13})$ of the the 13 missing observations (units > 4).

- Since y is unobserved, we must sample it, based on the given data.

- The joint posterior distribution of $\boldsymbol{y}$ and λ is

$$\begin{aligned}
&\pi\left(\lambda, y_1, \ldots, y_{13} \mid \boldsymbol{x}\right) \propto e^{-347\lambda} \lambda^{313} \left(\prod_{i=1}^{13} \lambda^{y_i} e^{-\lambda} \right) \frac{1}{\lambda} \\
&= e^{-347\lambda} \lambda^{313} \lambda^{\sum_{i=1}^{13} y_i} e^{-13\lambda} \frac{1}{\lambda} \\
&= \lambda^{313-1+\sum_{i=1}^{13} y_i} e^{-360\lambda}.
\end{aligned}$$

- FCD - $\lambda \mid \boldsymbol{y} \sim \mathcal{G}\left(313 + \sum_{i=1}^{13} y_i, 360\right)$
  - For $i = 1, \ldots, 13, y_i \mid \lambda \sim \mathcal{P}(\lambda)\mathbf{1}_{y_i \geq 4}$. This truncated Poisson may be sampled using rejection sampling with, for example, a Poisson or exponential envelope or using the while statement.

- - The posterior mean of λ is

$$\mathbb{E}[\lambda \mid \boldsymbol{y}] = \frac{313 + \sum_{i=1}^{13} y_i}{360}.$$

  - We can estimate λ using the Rao-Blackwellized estimator:

$$\begin{aligned}
\delta_{rb} &= \frac{1}{T} \sum_{t=1}^{T} \mathbb{E}\left[\lambda \mid y_1^{(t)}, \ldots, y_{13}^{(t)}\right] \\
&= \frac{1}{360T} \sum_{t=1}^{T} \left(313 + \sum_{i=1}^{13} y_i^{(t)}\right)
\end{aligned}$$

  where T is the number of post-burn in iterations.

- This provides a better estimator than using the λ^t directly.

- See hidden-poiss-data.R

Example 2.4.8

- The following table contains the number of failures of ten pumps in a nuclear plant (source: Gaver and O'Muircheartaigh, 1987).

Pump	1	2	3	4	5	6	7	8	9	10
Failures	5	1	5	14	3	19	1	1	4	22
Times	94.32	15.72	62.88	125.76	5.24	31.44	1.05	1.05	2.10	10.48

- These data can be modeled by assuming that the number of failures in the ith pump follows a Poisson process with parameter $\lambda_i (1 \leq i \leq 10)$.
- For any observation time t_i, the number of failures X_i is thus a Poisson $\mathcal{P}\left(\lambda_i t_i\right)$ random variable.
- We choose as priors for the λ_i a Gamma with parameters α and β.
- The prior on the hyperparameter β is also Gamma with parameters γ and δ.

Solution:

1. Compute the joint distribution $f(x, \lambda, \beta)$

$$f(x, \lambda, \beta) = f(x \mid \lambda, \beta) \times \pi(\lambda, \beta) = f(x \mid \lambda, \beta) \times \prod_{i=1}^{n} \pi(\lambda_i, \beta)$$

$$\propto \prod_{i=1}^{n} e^{-\lambda_i t_i} \frac{(\lambda_i t_i)^{x_i}}{x_i!} \times \prod_{i=1}^{n} \left(\lambda_i^{\alpha-1} e^{-\beta\lambda_i} \times \beta^{\gamma-1} e^{-\delta\beta}\right)$$

2. Deduce the FCD of λ_i $(i = \overline{1, 10})$ and β.

- FCD of λ_i

$$f(\lambda \mid x, \beta) \propto \prod_{i=1}^{10} e^{-\lambda_i t_i - \beta\lambda_i} \lambda_i^{x_i+\alpha-1} = \prod_{i=1}^{10} e^{-(t_i+\beta)\lambda_i} \lambda_i^{x_i+\alpha-1}$$

Hence $\lambda_i \mid x, \beta \sim \text{Gamma}\left(x_i + \alpha, t_i + \beta\right)$

- FCD of β

$$f(\beta \mid x, \lambda) \propto e^{-\beta \sum_{i=1}^{10} \lambda_i} \times \beta^{10\gamma-10} e^{-10\delta\beta} = \beta^{10\gamma-9-1} e^{-(\sum_{i=1}^{10} \lambda_i + 10\delta)\beta}$$

Hence $\beta \mid x, \lambda \sim \text{Gamma}(10\gamma - 9, \sum_{i=1}^{10} \lambda_i + 10\delta)$

Example 2.4.9

- The Failures-times program implements the Gibbs sampler.
- The hyperparameter values are $\alpha = 1.8, \gamma = 0.01$ and $\delta = 1$
- The total number of iterations is 5000 and the burn-in period is 500.
- Histograms of marginal distributions of β, λ_1 and λ_2 from these pump failure data are plotted.
- The estimates are for these three parameters $\hat{\beta} = 2.473$, $\hat{\lambda}_1 = 0.07$ and $\hat{\lambda}_2 = 0.155$.

Chapter 3

Image

3.1 What is an image?

- A digital image is composed of picture elements called pixels.
- Each pixel is assigned an intensity, meant to characterize the color of a small rectangular segment of the scene.
- A small image typically has around $256^2 = 65536$ pixels; while a high-resolution image often has 5 to 10 million pixels.

3.2 How image become an array of numbers

- Having a way to represent images as arrays of numbers is crucial to processing images using mathematical techniques.
- If you enter the 9×16 array into a MATLAB variable X (see code), and display the array with the commands imagesc (X), axis image, colormap (gray), then we obtain the first picture shown.
- The entries with value 8 are displayed as white, entries equal to zero are black, and values in between are shades of gray.
- Color images can be represented using various formats.
- The RGB format stores images as three components, which represent their intensities on the red, green, and blue scales.
- A pure red color is represented by the intensity values $(1, 0, 0)$ and $(0, 0, 1)$ represent pure blue.
- Other colors can be obtained with different choices of intensities. For example, the values $(1, 1, 0)$ represent yellow.
- Hence, to represent a color image, we need three values per pixel.
- For example, if X is a multidimensional MATLAB array of dimensions $9 \times 16 \times 3$, then we can display this image, in color, with the command imagesc (X), obtaining the second picture shown (cf code).

- A digital image is a two or three - dimensional array of numbers representing intensities on a grayscale or color scale.

3.3 Why do we need to restore/enhance an image?

Image can suffer several degradations due to

- Acquisition system components (resolution).
- Motion blur, focus blur.
  - The optical system in a camera lens may be out of focus, so that the incoming light is smeared out.
  - The same problem arises, for example, in astronomical imaging where the incoming light in the telescope has been slightly bent by turbulence in the atmosphere.
  - In these and similar situations, the inevitable result is that we record a blurred image.
- Ravages of time.
- etc.

Remark 3.3.1

Image processing may aim at: Noise reduction, Blur reduction, Extract reduction, Restoration...

Image enhancement

Remark 3.3.2

- It is used in the restoration of older movies
- For example, the original Star Wars trilogy was enhanced for release on DVD.
- These methods are not model-based and therefore not covered here.

Image restoration

Remark 3.3.3

- It is used in the restoration of older movies
- For example, the original Star Wars trilogy was enhanced for release on DVD.
- These methods are not model-based and therefore not covered here.

Remark 3.3.4

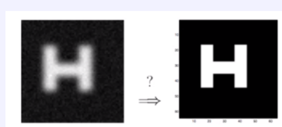
- In contrast to image enhancement, in image restoration we seek to recover the original, by using a mathematical model of the blurring process. Unfortunately, there is no hope that we can recover the original image exactly! This is due to various unavoidable errors in the recorded image.
- The most important errors are :
  - Fluctuations in the recording process.
  - Approximation errors when representing the image with a limited number of digits.
  - The influence of this noise puts a limit on the size of the details that we can hope to recover in the reconstructed image, and the limit depends on both the noise and the blurring process.

Remark 3.3.5

- In practice, when processing images, we rearrange their elements into column vectors by stacking the columns of these images into long vectors. The mathematical notation for this operator is vec.
- Example: If X is a $n \times m$ image, we define the vector $\mathbf{x} = \mathrm{vec}(X)$, of length $N = mn$: $x = (x_1, \ldots, x_N)^T$.

Example 3.3.1: Example of image restoration (Denoising)

- g measured image (noised) and f the desired image (denoised).
- Observation model:
  - Additive Noise $g(x, y) = f(x, y) + b(x, y)$.



  - An observed image can often be modeled as (if g and h are continuous functions) :

$$
\begin{aligned}
g(x, y) &= f(x, y) \otimes h(x, y) + b(x, y) \\
&= \iint_D f\left(x', y'\right) h\left(x - x', y - y'\right) \mathrm{d}x' \mathrm{d}y' + b(x, y) \\
&= \iint_D h\left(x', y'\right) f\left(x - x', y - y'\right) \mathrm{d}x' \mathrm{d}y' + b(x, y)
\end{aligned}
$$

By convolving the true image with the PSF, we obtain a blurred image. Each value of the function g is essentially a weighted average of the values of f where the weights are given by the function h.

Example 3.3.2: Example of image restoration

Deconvolution

- The discrete version of convolution is a summation over a finite number of terms.
- That is, pixels of the blurred image are obtained from a weighted sum of the corresponding pixel and its neighbors in the true image.
- The weights are given by the elements in the PSF array (see later).
- Observation model :

$$g(m,n) = \sum_{k=-P}^{P} \sum_{l=-Q}^{Q} h(k,I) f(m-k, n-l) + b(m,n),$$

- Convolution can be written as a matrix-vector multiplication :

$$\boldsymbol{g} = \boldsymbol{h} * \boldsymbol{f} + \boldsymbol{b} = \boldsymbol{H}\boldsymbol{f} + \boldsymbol{b}.$$

- g observed image ; f the desired image (deconvoluted) and h the point spread function of the imaging system, b additive noise.
- The objective of image restoration, in this case, is to estimate the original image f from the observed degraded image g (and h).
- If $\boldsymbol{H}$ is known : restoration = deconvolution.
- If $\boldsymbol{H} = \mathbf{I}$: restoration = denoising.
- If $\boldsymbol{H}$ partially known: restoration = blind or myopic decopvolution.

3.4 Deconvolution

Definition 4.1

- f: image before degradation (true image),
- g: image after degradation (observed image),
- h: point spread function (PSF) or impulse response (degradation filter),
- b: additive noise.

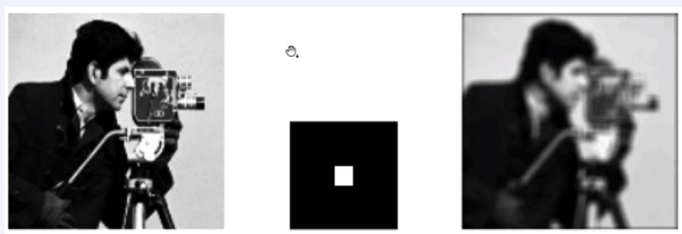
$\square$

Remark 3.4.1

- Suppose we take the exact image to be all black, except for a single bright pixel.
- If we take a picture of this image, then the blurring operation will cause the single bright pixel to be spread over its neighboring pixels.
- The single bright pixel is called a point source, and the function that describes the blurring and the resulting image of the point source is called the point spread function (PSF).

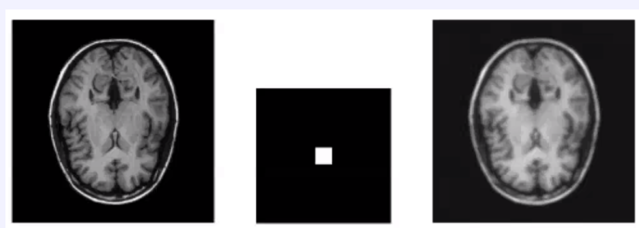
Example 3.4.1

$$g = h * f$$



True $\boldsymbol{f}$, impulse response $\boldsymbol{h}$ and observation $\boldsymbol{g}$.

Blurring acts as a low pass filter and attenuates higher spatial frequencies.



True $\boldsymbol{f}$, impulse response $\boldsymbol{h}$ and observation $\boldsymbol{g}$.

- One of the challenges of image deblurring is to devise efficient and reliable algorithms for recovering as much information as possible from the given data.
- We are now going to provide a brief introduction to the basic image deblurring problem and explain why it is difficult.

3.4.1 1D Deconvolution

- To fix the ideas, we consider the problem in 1D, i.e. signal deconvolution.
- Goal: understand the concepts and ideas before extending to 2D.
- "The most important thing is to understand the simple things, the complicated things are easy to deal with", Michel Talagrand.

- Convolutive Model (discrete 1D) :

$$g_n = \sum_{p=-P}^{P} h_p f_{n-p} + b_n.$$

- If we have N samples,

$$\begin{aligned}
&\vdots \\
&g_{n-1} = \dots \\
&g_n = h_P f_{n-P} + \dots + h_1 f_{n-10!} h_0 f_n + h_{-1} f_{n+1} + \dots + h_{-P} f_{n+P} \\
&\vdots \\
&g_{n+1} = \dots \\
&\vdots
\end{aligned}$$

- We can write these N equations in matrix form:

$$g = \boldsymbol{H} f + \boldsymbol{b}$$

- $\boldsymbol{g}$: N-dimensional vector of observations (in 2D, this will be the blurred image),
- $\boldsymbol{H}$: convolution matrix with size $N \times N$. It has the following form (Toeplitz, with the zero boundaries) :
- f: variable of interest (restored signal),
- $\boldsymbol{b}$: model errors and noise.

Remark 3.4.2: Deconvolution (inverse problem)

- Aim : from the observed data $\boldsymbol{g}$, find $\boldsymbol{f}$ assuming $\boldsymbol{H}$ is known.
- Note: $\boldsymbol{H}$ is often only partially known (myopic).

Definition 4.2

- Simplified case : we ignore the noise, $\boldsymbol{b} = 0$ and we suppose that $\boldsymbol{H}$ is invertible.
- The solution is then :

$$\hat{f} = \boldsymbol{H}^{-1} g$$

□

Difficulty

- Now supposes that the output $\boldsymbol{g}$ output has been measured in an imperfect form, for example, without measurement error, but in a uniformly quantized way on 10 bits, that is 1024 levels.

- We denote $\boldsymbol{g}_q$ this quantized output.
- The difference between $\boldsymbol{g}$ and $\boldsymbol{g}_q$ is imperceptible...
- However, the deconvolution image is far from f.

Example 3.4.2

- We consider as input (true image) f of size 256×256.
- The output (the data) is a blurred image $\boldsymbol{g}$ of the same size.
- Goal: find f.

3.4.2 2D Deconvolution

We slightly noise the g data, by adding some Gaussian noise with 0 mean and standard deviation 10^{-5}.
- The difference between $\boldsymbol{g}$ and the noisy version is imperceptible.
- ... however, the deconvoluted image has any feature of f!

Remark 3.4.3

- Why does the naive approach fail?
  1. The $\boldsymbol{H}$ matrix is often ill-conditioned (small error on $\boldsymbol{g}$ implies large errors on f (example 1).
  2. Furthermore, because the observations are collected by a mechanical device, there are (always) random perturbations in the recorded data $\hookrightarrow$ noise.
  3. Moreover, when the image is digitized, it is represented by a finite (and typically small) number of digits.
- Thus the recorded blurred image g is really given by:

$$\boldsymbol{g} = \boldsymbol{H}\boldsymbol{f} + \boldsymbol{b}$$

  where the noise image b (of the same dimension as g) represents the noise and the discretization errors in the recorded image.

Definition 4.3

- The naive solution therefore computes:

$$\hat{f} = \boldsymbol{H}^{-1}\boldsymbol{g} = \boldsymbol{H}^{-1}(\boldsymbol{H}\boldsymbol{f} + \boldsymbol{b}) = \boldsymbol{f} + \boldsymbol{H}^{-1}b.$$

- Two terms : the exact image $\boldsymbol{f}$ and the inverted noise $H^{-1}b$.
- If the image solution is unacceptable here, it is because the inverted noise strongly contaminates the reconstructed image due to the ill-conditioning nature of $\boldsymbol{H}$.
- Apparently, **image deblurring is not as simple as it first appears**.

□

Definition 4.4: Singular value decomposition

- The SVD of a square matrix $\boldsymbol{H} \in \mathbb{R}^{N \times N}$ is defined as the decomposition :

$$\boldsymbol{H} = \mathbf{U\Sigma V}^\top,$$

- $\mathbf{U}$ and $\mathbf{V}$ are orthogonal matrices satisfying $\mathbf{U}^T\mathbf{U} = I_N$ and $\mathbf{V}^T\mathbf{V} = I_N$.
- $\mathbf{\Sigma} = \mathrm{diag}\,(\sigma_i)$ is a diagonal matrix whose elements σ_i are nonnegative and appear in nonincreasing order

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_N \geq 0.$$

- The quantities σ_i are called the singular values, and the rank of $\boldsymbol{H}$ is equal to the number of positive singular values.
- Assuming for the moment that all singular values are strictly positive, it is straightforward to show that the inverse of $\boldsymbol{H}$ is given by :

$$\boldsymbol{H}^{-1} = \mathbf{V\Sigma}^{-1}\mathbf{U}^\top.$$

- Since $\mathbf{\Sigma}$ is a diagonal matrix, its inverse $\mathbf{\Sigma}^{-1}$ is also diagonal, with entries $1/\sigma_i$ for $i = 1, \ldots, N$.

□

- Another representation of $\boldsymbol{H}$ and $\boldsymbol{H}^{-1}$ that is also useful to us:

$$\boldsymbol{H} = \mathbf{U\Sigma V}^\top = \sum_{i=1}^{N} \sigma_i \boldsymbol{u}_i \boldsymbol{v}_i^T$$

- Similarly,

$$\boldsymbol{H}^{-1} = \sum_{i=1}^{N} \frac{1}{\sigma_i} \boldsymbol{v}_i \boldsymbol{u}_i^T$$

  where $\boldsymbol{u}_i$ (resp. $\boldsymbol{v}_i$) is the i-th column of the matrix $\mathbf{U}$ (resp. $\mathbf{V}$).

- The naive solution can be written as :

$$\hat{\boldsymbol{f}} = \boldsymbol{H}^{-1}\boldsymbol{g} = \mathbf{V\Sigma}^{-1}\mathbf{U}^\top\boldsymbol{g} = \sum_{i=1}^{N} \frac{\boldsymbol{u}_i^T\boldsymbol{g}}{\sigma_i}\boldsymbol{v}_i$$

- And the inverted noise contribution to the solution is given by

$$\boldsymbol{H}^{-1}\boldsymbol{b} = \mathbf{V\Sigma}^{-1}\mathbf{U}^\top\boldsymbol{b} = \sum_{i=1}^{N} \frac{\boldsymbol{u}_i^T\boldsymbol{b}}{\sigma_i^\star}\boldsymbol{v_i}$$

- For image deblurring problems, the decay of the singular values to a value very close to zero.

- As a consequence the condition number :

$$\mathrm{cond}(\boldsymbol{H}) = \sigma_1/\sigma_N$$

is very large, indicating that the solution is very sensitive to perturbation and rounding errors.

- Looking at the expression $\boldsymbol{H}^{-1}\boldsymbol{b}$ we see that the quantities $\dfrac{\boldsymbol{u}_i^\top \boldsymbol{b}}{\sigma_i}$, are the expansion coefficients for the basis vectors $\boldsymbol{v_i}$.

- When these quantities are small in magnitude, the solution has very little contribution from $\boldsymbol{v}_i$.

- But when we divide by a small singular value such as σ_N, we considerably amplify the corresponding component, $\boldsymbol{u}_N^T\boldsymbol{b}$, which in turn contributes a large multiple of the high-frequency information contained in $\boldsymbol{v}_N$ to the computed solution.

- **Note: the singular vectors corresponding to the smaller singular values typically represent higher-frequency information.**

- This is precisely why a naive reconstruction, such as the one in the previous figure, appears as a random image dominated by high frequencies.

Remark 3.4.4

- For this reason, one can think of ignoring the high-frequency components, as they are dominated by error.
- **The truncated SVD algorithms** use this idea.

Remark 3.4.5: About the PSF

- In some cases the PSF can be described analytically, and thus the PSF array can be constructed from a function.
- Knowledge of the physical process that causes the blur provides an explicit formulation of the PSF.
- **Example:** The PSF for blurring caused by atmospheric turbulence can be described as a 2D Gaussian function.
- The blurring matrix $\boldsymbol{H}$ is determined from two ingredients: the PSF, which defines how each pixel is blurred, and the boundary conditions, which specify our assumptions on the scene just outside our image.

Remark 3.4.6: Boundary conditions

- Recall that g_{ij} is obtained from a weighted sum of pixel f_{ij} and its neighbors.
- Some of these neighbors may be outside the field of view.
- A model for image deblurring must take account of these boundary effects.
- The simplest boundary condition is to assume that the exact image is black (i.e., consists of zeros) outside the boundary.
- The **zero boundary condition** is a good choice when the exact image is mostly zero outside the boundary. Example: astronomical images.
- The **periodic boundary condition** is frequently used in image processing. This implies that the image repeats itself (endlessly) in all directions.

3.4.3 Some approaches to deconvolution

1D Deconvolution

- Recall :

$$\boldsymbol{g} = \boldsymbol{H}\boldsymbol{f} + \boldsymbol{b}.$$

- $\boldsymbol{H}$ is not necessarily invertible and there is noise.

- Least squares (LS) solution :

$$\hat{\boldsymbol{f}}_{LS} = \arg\min_{\boldsymbol{f}} \|\boldsymbol{g} - \boldsymbol{H}\boldsymbol{f}\|^2.$$

- Criterion to be minimized:

$$\Phi_{LS}(\boldsymbol{f}) = \|\boldsymbol{g} - \boldsymbol{H}\boldsymbol{f}\|^2$$

- Calculating the solution :

$$\begin{aligned}
&\Phi_{LS}(\boldsymbol{f}) = \langle \boldsymbol{g} - \boldsymbol{H}\boldsymbol{f}, \boldsymbol{g} - \boldsymbol{H}\boldsymbol{f} \rangle \\
&= \langle \boldsymbol{g}, \boldsymbol{g} \rangle - 2\langle \boldsymbol{g}, \boldsymbol{H}\boldsymbol{f} \rangle + \langle \boldsymbol{H}\boldsymbol{f}, \boldsymbol{H}\boldsymbol{f} \rangle \\
&= \boldsymbol{g}^T\boldsymbol{g} - 2\boldsymbol{g}^T\boldsymbol{H}\boldsymbol{f} + \boldsymbol{f}^T\boldsymbol{H}^T\boldsymbol{H}\boldsymbol{f} \\
&\frac{\partial \Phi_{LS}}{\partial \boldsymbol{f}} = 0 \Leftrightarrow -2\boldsymbol{H}^T\boldsymbol{g} + 2\boldsymbol{H}^T\boldsymbol{H}\boldsymbol{f} = 0 \Leftrightarrow \boldsymbol{H}^T\boldsymbol{H}\boldsymbol{f} = \boldsymbol{H}^T\boldsymbol{g} \\
&\hat{\boldsymbol{f}}_{LS} = \left(\boldsymbol{H}^T\boldsymbol{H}\right)^{-1}\boldsymbol{H}^T\boldsymbol{g}.
\end{aligned}$$

Example 3.4.3: inversion of large size matrices

- Matrix $\boldsymbol{H}^T\boldsymbol{H}$ of size $\boldsymbol{N} \times \boldsymbol{N}$.
- For an 1000×1000 image, $N = 10^6$ and then $\boldsymbol{H}^T\boldsymbol{H}$ is of size $10^6 \times 10^6$.
- Much more in 3D!
- Inversion is time-consuming or impossible.
- Solutions for inverting these large matrices :
  1. Numerical optimization (gradient algorithms).
  2. Circulant Approximation (Hunt, 1970).
- Very common approximation: assume that the signal to restore is N-periodic.
- The convolution matrix is the a circulant matrix denoted $\tilde{\boldsymbol{H}}$:

Definition 4.5: Circulant matrix

- **A circulant matrix** is a square matrix in which all row vectors are composed of the same elements and each row vector is rotated one element to the right relative to the preceding row vector.
- A $N \times N$ circulant matrix takes the form :

$$\boldsymbol{C} = \begin{pmatrix} c_1 & c_2 & c_3 & \cdots & c_2 \\ c_N & c_1 & c_2 & & c_{N-1} \\ c_{N-1} & c_N & c_1 & & c_{N-2} \\ \vdots & & & \ddots & \vdots \\ c_2 & c_3 & c_4 & \cdots & c_1 \end{pmatrix}$$

□

- This matrix can be diagonalized...

$$\tilde{\boldsymbol{H}} = \boldsymbol{F}^*\boldsymbol{\Lambda}_h\boldsymbol{F},$$

- ... in the Fourier basis :

$$\boldsymbol{F} = N^{-1/2}\left[e^{-2i\pi(k-1)(I-1)/N}\right]_{k,l\in 1,\dots,N}.$$

- Reminder of the properties of $\boldsymbol{F}$:

$$\begin{aligned} \boldsymbol{F}^T &= F \quad \text{(symmetry)} \\ \boldsymbol{F}^*\boldsymbol{F} = \boldsymbol{F}\boldsymbol{F}^* &= \boldsymbol{I}_N \quad \text{(unitary)} \\ \boldsymbol{F}^{-1} &= \boldsymbol{F}^* \\ \boldsymbol{F}\boldsymbol{x} &= \mathrm{FFT}(\boldsymbol{x}) \\ \boldsymbol{F}^*\boldsymbol{x} &= \mathrm{IFFT}(\boldsymbol{x}) \end{aligned}$$

- Model : $\boldsymbol{g} = \boldsymbol{H}\boldsymbol{f} + \boldsymbol{b}$
- If $\boldsymbol{H}$ can be approximated by a circulant matrix $\tilde{\boldsymbol{H}}$, and letting $\hat{f} := \hat{\boldsymbol{f}}_{LS}$, we have:

$$\begin{aligned}
\hat{\boldsymbol{f}} &= \left(\boldsymbol{H}^T\boldsymbol{H}\right)^{-1}\boldsymbol{H}^T\boldsymbol{g} \\
&\approx \left(\tilde{\boldsymbol{H}}^T\tilde{\boldsymbol{H}}\right)^{-1}\tilde{\boldsymbol{H}}^T\boldsymbol{g} \\
&= \left[\boldsymbol{F}^*\boldsymbol{\Lambda}_h^*\boldsymbol{F}\boldsymbol{F}^*\boldsymbol{\Lambda}_h\boldsymbol{F}\right]^{-1}\left(\boldsymbol{F}^*\boldsymbol{\Lambda}_h^*\boldsymbol{F}\right)\boldsymbol{g} \text{ since } \tilde{\boldsymbol{H}}^T = \boldsymbol{F}^*\boldsymbol{\Lambda}_h^*\boldsymbol{F} \\
&= \left[\boldsymbol{F}^*\boldsymbol{\Lambda}_h^*\boldsymbol{\Lambda}_h\boldsymbol{F}\right]^{-1}\left(\boldsymbol{F}^*\boldsymbol{\Lambda}_h^*\boldsymbol{F}\right)\boldsymbol{g} \text{ since } \boldsymbol{F}\boldsymbol{F}^* = \boldsymbol{I} \\
&= \left(\boldsymbol{\Lambda}_h\boldsymbol{F}\right)^{-1}\left(\boldsymbol{F}^*\boldsymbol{\Lambda}_h^*\right)^{-1}\left(\boldsymbol{F}^*\boldsymbol{\Lambda}_h^*\right)\boldsymbol{F}\boldsymbol{g} \quad \text{red term } = \text{ id} \\
&= \boldsymbol{F}^{-1}\boldsymbol{\Lambda}_h^{-1}\boldsymbol{F}\boldsymbol{g} = \boldsymbol{F}^*\boldsymbol{\Lambda}_h^{-1}\boldsymbol{F}\boldsymbol{g} \quad \text{since } \boldsymbol{F} \text{ is unitary.}
\end{aligned}$$

- Multiplying by $\boldsymbol{F}$, we have :

$$\boldsymbol{F}\hat{\boldsymbol{f}} = \boldsymbol{F}\boldsymbol{F}^*\boldsymbol{\Lambda}_h^{-1}\boldsymbol{F}\boldsymbol{g} = \boldsymbol{\Lambda}_h^{-1}\boldsymbol{F}\boldsymbol{g}$$

- Since $\boldsymbol{F}\hat{\boldsymbol{f}} = \widehat{\boldsymbol{f}}$ and $\boldsymbol{F}\boldsymbol{g} = \mathring{\mathrm{g}}$, we have

$$\widehat{\boldsymbol{f}} = \boldsymbol{\Lambda}_h^{-1}\,\mathring{\boldsymbol{g}}$$

Remark 3.4.7

- Why is the deconvoluted image worse than the observations?
- Recall: for $n = 1, \ldots, N$,
- Division by small values of $\mathring{n}_n$ yields very large distortions
- Such noise amplification produces large artifacts displayed by the solution, which is unacceptable.
- **Advantages and drawbacks:**
  1. Very general, fast, no parameters to tune.
  2. ... but does not work in general!
  3. Can produce unacceptable, explosive solutions.

Penealized Least square

- LS solutions; minimize the energy of the error between the model $\boldsymbol{H}\boldsymbol{f}$ and the data g.
- Therefore, they provide, in this sense, high fidelity to the data.
- Problem: the components of the restored object may have high amplitudes resulting mainly from noise.
- $\Rightarrow$ LS solutions are then unacceptable.
- Solution: introduce some infidelity to the data.
  Aim: obtain a solution less rough than the LS one and more in accordance with the idea we have a priori.

- $\Rightarrow$ Minimization of a composite criterion.
- $\hookrightarrow$ Take into account a prior information on $\boldsymbol{f}$.
- Prior on the smoothness, the regularity etc.
- Regularization by penalizing : we minimize

$$\Phi_{\mathrm{PLS}}(\boldsymbol{f}) = \|\boldsymbol{g} - \boldsymbol{H}\boldsymbol{f}\|^2 + \mu\mathcal{P}(\boldsymbol{f})$$

  where $\mathcal{P}(\boldsymbol{f})$ is a constraint operator.

- Restored Signal :

$$\hat{f}_{\mathrm{PLS}} = \arg\min_{\boldsymbol{f}} \Phi_{PLS}(\boldsymbol{f}).$$

- Terms in Φ_{PLS} :
  1. $\|\boldsymbol{g} - \boldsymbol{H}\boldsymbol{f}\|^2$: fit to the data $\hookrightarrow$ Term used to quantify the similarity between the observed signal and the reconstructed signal.
  2. $\mathcal{P}(f)$: regularization term. $\hookrightarrow$ Ensures that the restored signal has some regularity.
  3. μ : regularization parameter. $\hookrightarrow$ A compromise between the two....
- Regularization : a compromise...
- Criterion :

$$\Phi_{\mathrm{PLS}}(\boldsymbol{f}) = \|\boldsymbol{g} - \boldsymbol{H}\boldsymbol{f}\|^2 + \mu\mathcal{P}(\boldsymbol{f}).$$

- If $\mu \to 0, \hat{\boldsymbol{f}} \to \arg\min \boldsymbol{f}\|\boldsymbol{g} - \boldsymbol{H}\boldsymbol{f}\|^2$: perfect fidelity to the data, non-regularized LS solution..
- If $\mu \to +\infty, \hat{f} \to \arg\min f\mathcal{P}(\boldsymbol{f})$: fidelity to the prior, solution not depending on $\boldsymbol{g}$.

Choice of $\mathcal{P}(\boldsymbol{f})$

- We want the regularization term to be small when f has the criteria we expect from a good solution.
- For example, we want to penalize the too-fast variations $\rightarrow$ smoothness.
- Smooth function: penalize the derivatives.
- The justification is that differentiation is known to magnify high-frequency components.
- $\rightarrow$ We can often ensure the computation of a smooth solution by controlling the size of the partial derivatives.
- Differences, higher order derivatives, generalizations :

$$\mathcal{P}(\boldsymbol{f}) = \sum_n (f_{n+1} - f_n)^2$$
$$\mathcal{P}(\boldsymbol{f}) = \sum_n (f_{n+1} - 2f_n + f_{n-1})^2$$
$$\mathcal{P}(\boldsymbol{f}) = \sum_n (\alpha f_{n+1} - f_n + \alpha' f_{n-1})^2$$

- 2D aspects: differences between neighboring pixels,

$$\begin{aligned}\mathcal{P}(\boldsymbol{f}) &= \sum_{p\sim q} (f_p - f_q)^2 \\ &= \sum_{n,m} (f_{n+1,m} - f_{n,m})^2 + \sum_{n,m} (f_{n,m+1} - f_{n,m})^2\end{aligned}$$

- Norms of filtered images:
  1. Filtering rows and columns :
  $$\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} + \begin{bmatrix} 1 & 0 & -1 \end{bmatrix}$$
  2. Unique filter (laplacian)
  $$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$
  3. Any highpass filter, contour detector (Prewitt, Sobel, etc.).
  4. Other possibilities:
     (a) Reinforcement towards a known form f_0 :
     $$\mathcal{P}(\boldsymbol{f}) = (\boldsymbol{f} - \boldsymbol{f}_0)^T (\boldsymbol{f} - \boldsymbol{f}_0).$$
     (b) Usual euclidean norm (separable terms) :
     $$\mathcal{P}(\boldsymbol{f}) = \boldsymbol{f}\boldsymbol{f}^T.$$
- More general form :
$$\mathcal{P}(\boldsymbol{f}) = (\boldsymbol{f} - \boldsymbol{f}_0)^T M (\boldsymbol{f} - \boldsymbol{f}_0)$$
- In the following development (Hunt-Wiener model).
- In 1D
$$\mathcal{P}(\boldsymbol{f}) = \sum_n (f_{n+1} - f_n)^2 = \|\boldsymbol{D}\boldsymbol{f}\|^2 = \boldsymbol{f}^T \boldsymbol{D}^T \boldsymbol{D}\boldsymbol{f},$$

  where $\boldsymbol{D} = \dots$ ... where $\boldsymbol{D}$ is the difference matrix, of size $N \times N$:

$$D = \begin{pmatrix} 1 & -1 & 0 & 0 & \dots & \dots & 0 \\ 0 & 1 & -1 & 0 & \cdots & \dots & 0 \\ \vdots & & & \ddots & \ddots & & \vdots \\ \vdots & & & & \ddots & \ddots & \vdots \\ 0 & \dots & \dots & 0 & 1 & -1 & 0 \\ 0 & \dots & \dots & 0 & 0 & 1 & -1 \\ -1 & \cdots & \dots & 0 & 0 & 0 & 1 \end{pmatrix}$$

- Note: $\boldsymbol{D}$ is circulant.

- Remind the criterion :

$$\begin{aligned}\Phi_{\text{PLS}}(\boldsymbol{f}) &= \|\boldsymbol{g}-\boldsymbol{H}\boldsymbol{f}\|^2+\mu\|\boldsymbol{D}\boldsymbol{f}\|^2\\ &= \langle \boldsymbol{g}-\boldsymbol{H}\boldsymbol{f}, \boldsymbol{g}-\boldsymbol{H}\boldsymbol{f}\rangle+\mu\langle \boldsymbol{D}\boldsymbol{f}, \boldsymbol{D}\boldsymbol{f}\rangle\\ &= \langle \boldsymbol{g}, \boldsymbol{g}\rangle-2\langle \boldsymbol{g}, \boldsymbol{H}\boldsymbol{f}\rangle+\langle \boldsymbol{H}\boldsymbol{f}, \boldsymbol{H}\boldsymbol{f}\rangle+\mu\langle \boldsymbol{D}\boldsymbol{f}, \boldsymbol{D}\boldsymbol{f}\rangle\\ &= \boldsymbol{g}^T\boldsymbol{g}-2\boldsymbol{g}^T\boldsymbol{H}\boldsymbol{f}+\boldsymbol{f}^T\boldsymbol{H}^T\boldsymbol{H}\boldsymbol{f}+\mu\boldsymbol{f}^T\boldsymbol{D}^T\boldsymbol{D}\boldsymbol{f}\end{aligned}$$

- Compute its gradient, Hessian, and the PLS solution

$$\begin{aligned}\nabla\Phi_{\text{PLS}}(f) &= 2H^T(Hf-g)+2\mu D^T(Df)\\ \nabla^2\Phi_{\text{PLS}}(f) &= 2H^TH+2\mu D^TD.\end{aligned}$$

- Normal system of equation :

$$\boldsymbol{H}^T\boldsymbol{g}=\boldsymbol{H}^T\boldsymbol{H}\boldsymbol{f}+\mu\boldsymbol{D}^T\boldsymbol{D}\boldsymbol{f}.$$

- PLS solution :

$$\hat{f}_{PLS}=\left(\boldsymbol{H}^T\boldsymbol{H}+\mu\boldsymbol{D}^T\boldsymbol{D}\right)^{-1}\boldsymbol{H}^T\boldsymbol{g}.$$

Quadratic Penalty

- Since the data fit term is quadratic if the penalty is also quadratic, so is ϕ_{PLS}.
- Advantage: fast and explicit calculation of $\hat{f}_{PLS}$ (linear function of $\boldsymbol{g}$).
- This quadratic criterion is suitable for stationary and smooth signals (no abrupt variations).
- Existence of other criteria, using a function that grows slower than a parabola, to less penalize large variations.
- If the penalty is convex (e.g. $\mathcal{P}(\boldsymbol{f})=\|\boldsymbol{f}\|$), quadratic + convex $\to$ convex criteria.
- No explicit solution but convexity ensures the convergence of standard minimization techniques.
- If the penalty is not convex, the criterion is not convex.
  We use iterative techniques, for example, gradient descent. $\hookrightarrow$ Advantage: really detect discontinuities.
  $\hookrightarrow$ Disadvantage: instability and high computational cost (possibility of local minima).

Back to our problem.
We have seen that the PLS solution is given by :

$$\hat{\boldsymbol{f}}_{PLS}=\left(\boldsymbol{H}^T\boldsymbol{H}+\mu\boldsymbol{D}^T\boldsymbol{D}\right)^{-1}\boldsymbol{H}^T\boldsymbol{g}.$$

- Even if an explicit solution exists, we don't use it directly. Why?
- Inversion of very large matrices.
- Circulant approximation of $\boldsymbol{H}$ by $\tilde{H}$ and diagonalization in the Fourier basis (previously done, with LS) :

$$\boldsymbol{H}\approx\tilde{\boldsymbol{H}}=\boldsymbol{F}^*\boldsymbol{\Lambda}_h\boldsymbol{F},$$

- Diagonalization of $\boldsymbol{D}$ in the Fourier basis :

$$\boldsymbol{D} = \boldsymbol{F}^* \boldsymbol{\Lambda}_d \boldsymbol{F}$$

-

$$\begin{aligned}\hat{\boldsymbol{f}}_{PLS} &\approx \left(\tilde{\boldsymbol{H}}^T \tilde{\boldsymbol{H}} + \mu \boldsymbol{D}^T \boldsymbol{D}\right)^{-1} \tilde{\boldsymbol{H}}^T \boldsymbol{g} \\ &= \left[\boldsymbol{F}^* \boldsymbol{\Lambda}_h^* \boldsymbol{F} \boldsymbol{F}^* \boldsymbol{\Lambda}_h \boldsymbol{F} + \mu \boldsymbol{F}^* \boldsymbol{\Lambda}_d^* \boldsymbol{F} \boldsymbol{F}^* \boldsymbol{\Lambda}_d \boldsymbol{F}\right]^{-1} \boldsymbol{F}^* \boldsymbol{\Lambda}_h^* \boldsymbol{F} \boldsymbol{g} \\ &= \boldsymbol{F}^* \left(\boldsymbol{\Lambda}_h^* \boldsymbol{\Lambda}_h + \mu \boldsymbol{\Lambda}_d^* \boldsymbol{\Lambda}_d\right)^{-1} \boldsymbol{\Lambda}_h^* \boldsymbol{F} \boldsymbol{g}.\end{aligned}$$

- Multiplying by F :

$$\boldsymbol{F} \hat{\boldsymbol{f}} = \left(\boldsymbol{\Lambda}_h^* \boldsymbol{\Lambda}_h + \mu \boldsymbol{\Lambda}_d^* \boldsymbol{\Lambda}_d\right)^{-1} \boldsymbol{\Lambda}_h^* \boldsymbol{F} \boldsymbol{g}.$$

- Since $\boldsymbol{F} \hat{\boldsymbol{f}} = \overset{\circ}{\hat{f}}$ and $\boldsymbol{F} \boldsymbol{g} = \overset{\circ}{\boldsymbol{g}}$, hence

$$\overset{\circ}{\hat{\boldsymbol{f}}} = \left(\boldsymbol{\Lambda}_h^* \boldsymbol{\Lambda}_h + \mu \boldsymbol{\Lambda}_d^* \boldsymbol{\Lambda}_d\right)^{-1} \boldsymbol{\Lambda}_h^* \overset{\circ}{\boldsymbol{g}}$$

3.4.4 Noise and error modelling

- Recall :

$$\boldsymbol{g} = \boldsymbol{H} \boldsymbol{f} + \boldsymbol{b}.$$

- We probabilize the measurement and modekking errors : $\boldsymbol{b} \mid \boldsymbol{f} \sim$ proba. dist.
- These errors are assumed to be centered in the sense that:

$$\boldsymbol{E}[\boldsymbol{b} \mid \boldsymbol{f}] = 0.$$

- This implies that

$$E[g \mid f] = \boldsymbol{H} f.$$

$$\text{since } E[g \mid f] = E[(\boldsymbol{H}\boldsymbol{f} + \boldsymbol{b}) \mid \boldsymbol{f}] = \boldsymbol{E}[\boldsymbol{H}\boldsymbol{f} \mid \boldsymbol{f}] + \boldsymbol{E}[\boldsymbol{b} \mid \boldsymbol{f}] = \boldsymbol{H}\boldsymbol{f}.$$

- The two formulations below are equivalent :

$$\begin{cases} g = \boldsymbol{H}\boldsymbol{f} + \boldsymbol{b} \\ \boldsymbol{E}[\boldsymbol{b} | \boldsymbol{f}] = 0. \end{cases} \Longleftrightarrow \begin{cases} E[\boldsymbol{g} \mid \boldsymbol{f}] = \boldsymbol{H}\boldsymbol{f} \\ \boldsymbol{b} = \boldsymbol{g} - \boldsymbol{H}\boldsymbol{f} \end{cases}$$

- We also have for the covariance:

$$C_{b|f} = C_{g|f}.$$

Indeed,

$$\begin{aligned}\boldsymbol{C}_{\boldsymbol{b}|\boldsymbol{f}} &= \boldsymbol{E}\left[(\boldsymbol{b} - \boldsymbol{E}[\boldsymbol{b} \mid \boldsymbol{f}]) \cdot (\boldsymbol{b} - \boldsymbol{E}[\boldsymbol{b} \mid \boldsymbol{f}])^T \mid \boldsymbol{f}\right] \\ &= \boldsymbol{E}\left[\boldsymbol{b} \cdot \boldsymbol{b}^T \mid \boldsymbol{f}\right] \quad \text{since } \boldsymbol{E}[\boldsymbol{b} \mid \boldsymbol{f}] = 0 \\ &= \boldsymbol{E}\left[(\boldsymbol{g} - \boldsymbol{H}\boldsymbol{f}) \cdot (\boldsymbol{g} - \boldsymbol{H}\boldsymbol{f})^T \mid \boldsymbol{f}\right] \quad \text{because } \boldsymbol{b} = \boldsymbol{g} - \boldsymbol{H}\boldsymbol{f} \\ &= \boldsymbol{E}\left[(\boldsymbol{g} - \boldsymbol{E}[\boldsymbol{g} \mid \boldsymbol{f}]) \cdot (\boldsymbol{g} - \boldsymbol{E}[\boldsymbol{g} \mid \boldsymbol{f}])^T \mid \boldsymbol{f}\right] \quad \text{since } \boldsymbol{H}\boldsymbol{f} = \boldsymbol{E}[\boldsymbol{g} \mid \boldsymbol{f}] \\ &= \boldsymbol{C}_{\boldsymbol{g}|\boldsymbol{f}} \quad \text{by definition of covariance.}\end{aligned}$$

3.4.5 Gaussian Approximation of erros

- If we assume that the errors follow this normal distribution :

$$\boldsymbol{b} \mid \boldsymbol{f} \sim \mathcal{N}\left(\mathbf{0}, \boldsymbol{C}_{\boldsymbol{b}|\boldsymbol{f}}\right).$$

- The p.d.f is given by ;

$$p(\boldsymbol{b} \mid \boldsymbol{f}) = \frac{\exp\left(-\frac{1}{2}\boldsymbol{b}^T \cdot \boldsymbol{C}_{\boldsymbol{b}|\boldsymbol{f}}^{-1} \cdot \boldsymbol{b}\right)}{(2\pi)^{N/2}\left|\boldsymbol{C}_{\boldsymbol{b}|\boldsymbol{f}}\right|^{1/2}}$$

  where N is the number of measurements.

- We have then, for the data $\boldsymbol{g} \mid \boldsymbol{f} \sim \mathcal{N}\left(\boldsymbol{H}\boldsymbol{f}, \boldsymbol{C}_{\boldsymbol{g}|\boldsymbol{f}}\right)$, with p.d.f

$$\begin{aligned} p(\boldsymbol{g} \mid \boldsymbol{f}) &= \frac{\exp\left(-\frac{1}{2}(\boldsymbol{g} - \boldsymbol{H}\boldsymbol{f})^T \cdot \boldsymbol{C}_{\boldsymbol{g}|\boldsymbol{f}}^{-1} \cdot (\boldsymbol{g} - \boldsymbol{H}\boldsymbol{f})\right)}{(2\pi)^{N/2}\left|\boldsymbol{C}_{g|\boldsymbol{f}}\right|^{1/2}} \\ &= (2\pi)^{-N/2}\left|\boldsymbol{C}_{g|\boldsymbol{f}}\right|^{-1/2} \exp\left(-\frac{1}{2}\|\boldsymbol{g} - \boldsymbol{H}\boldsymbol{f}\|^2_{C^{-1}_{g|f}}\right). \end{aligned}$$

3.4.6 Maximum likelihood (ML)

- Using this distribution $p(\boldsymbol{g} \mid \boldsymbol{f})$, we define the solution of the inverse problem by the one maximizing the likelihood :
- Likelihood: $p(\boldsymbol{g} \mid \boldsymbol{f})$ viewed as a function of the parameter $\boldsymbol{f}$.
- ML solution :

$$\hat{\boldsymbol{f}}_{ML} = \arg\max_{\boldsymbol{f}} p(\boldsymbol{g} \mid \boldsymbol{f}).$$

- We instead maximize the log :

$$\begin{aligned} \hat{\boldsymbol{f}}_{ML} &= \arg\max_{\boldsymbol{f}} \ln[p(\boldsymbol{g} \mid \boldsymbol{f})] \\ &= \arg\max_{\boldsymbol{f}} \ln\left[\exp\left(-\frac{1}{2}\|\boldsymbol{g} - \boldsymbol{H}\boldsymbol{f}\|^2_{C^{-1}_{g|f}}\right)\right] \\ &= \arg\min_{\boldsymbol{f}} \|\boldsymbol{g} - \boldsymbol{H}\boldsymbol{f}\|^2_{C^{-1}_{g|f}}. \end{aligned}$$

- Penalty function :

$$\Phi_{g|f}(f) = (\boldsymbol{g} - \boldsymbol{H}\boldsymbol{f})^T \cdot C^{-1}_{g|f} \cdot (\boldsymbol{g} - \boldsymbol{H}\boldsymbol{f})$$

- 1st order optimality condition :

$$\left.\frac{\partial \Phi_{g|f}}{\partial \boldsymbol{f}}\right|_{\boldsymbol{f}=\boldsymbol{f}_{ML}} = 0$$

- Sufficient condition here.
- The ML solution is then :

$$\hat{\boldsymbol{f}}_{ML} = \left(\boldsymbol{H}^T \cdot \boldsymbol{C}^{-1}_{\boldsymbol{g}|\boldsymbol{f}} \cdot \boldsymbol{H}\right)^{-1} \cdot \boldsymbol{H}^T \cdot \boldsymbol{C}^{-1}_{\boldsymbol{g}|\boldsymbol{f}} \cdot \boldsymbol{g}.$$

- But the matrix $\boldsymbol{H}$ is often ill-conditioned.
- The resolution operator

$$\boldsymbol{H}^{\dagger} := \left(\boldsymbol{H}^T \cdot \boldsymbol{C}_{g|f}^{-1} \cdot \boldsymbol{H}\right)^{-1} \cdot \boldsymbol{H}^T \cdot \boldsymbol{C}_{\boldsymbol{g}|\boldsymbol{f}}^{-1},$$

  is unstable.
- The solution $\hat{\boldsymbol{f}}_{ML} = \boldsymbol{H}^{\dagger}\boldsymbol{g}$ is then unacceptable : the noise amplification is excessive.

Example 3.4.4: white Gaussian noise

- Let $\boldsymbol{b} \sim \mathcal{N}\left(\mathbf{0}, \sigma_b^2 \boldsymbol{I}\right)$, i.e $\forall i, b_i \sim \mathcal{N}\left(0, \sigma_b^2\right), b_i \perp b_j$.
- We have then for the data $g \mid \boldsymbol{f} \sim \mathcal{N}\left(\boldsymbol{H}\boldsymbol{f}, \sigma_b^2 I\right)$, with density

$$\begin{aligned} p(\boldsymbol{g} \mid \boldsymbol{f}) &\propto \exp\left(-\frac{1}{2\sigma_b^2}(\boldsymbol{g} - \boldsymbol{H}\boldsymbol{f})^T \cdot (\boldsymbol{g} - \boldsymbol{H}\boldsymbol{f})\right) \\ &= \exp\left(-\frac{1}{2\sigma_b^2}\|\boldsymbol{g} - \boldsymbol{H}\boldsymbol{f}\|^2\right). \end{aligned}$$

- We thus have :

$$\begin{aligned} \hat{f}_{ML} &= \arg\max_f \ln[p(\boldsymbol{g} \mid \boldsymbol{f})] \\ &= \arg\min_{\boldsymbol{f}} \|\boldsymbol{g} - \boldsymbol{H}\boldsymbol{f}\|^2 \\ &= \hat{\boldsymbol{f}}_{LS}. \end{aligned}$$

3.4.7 Regularization using Bayesian approach

Inverse ill-posed problem:

- Data is not informative enough: respecting the data is not enough.
- $\hookrightarrow$ Take into account a prior information on f through a probability distribution.
- Likelihood : $p(\boldsymbol{g} \mid \boldsymbol{f})$, unchanged.
- Prior on $\boldsymbol{f}$: $p(\boldsymbol{f})$
- Posterior of f :

$$p(\boldsymbol{f} \mid \boldsymbol{g}) = \frac{p(\boldsymbol{g} \mid \boldsymbol{f})p(\boldsymbol{f})}{p(\boldsymbol{g})}.$$

- Use $p(\boldsymbol{f} \mid \boldsymbol{g})$ to compute a solution.
- Examples :

$$\hat{\boldsymbol{f}}_{MAP} = \arg\max_{\boldsymbol{f}} p(\boldsymbol{f} \mid \boldsymbol{g}).$$

$$\hat{\boldsymbol{f}}_{PM} = \boldsymbol{E}[\boldsymbol{f} \mid \boldsymbol{g}] = \int \boldsymbol{f} p(\boldsymbol{f} \mid \boldsymbol{g}) d\boldsymbol{f}.$$

- And much more...

3.4.8 Gaussian Model

$$g = Hf + b.$$

- Likelihood : $\boldsymbol{g} \mid \boldsymbol{f} \sim \mathcal{N}\left(\boldsymbol{Hf}, \boldsymbol{C_{g|f}}\right)$.
- Prior: $f \sim \mathcal{N}\left(\bar{f}, C_f\right)$.
- We have seen previously that $\boldsymbol{g} \mid \boldsymbol{f} \sim \mathcal{N}\left(\boldsymbol{Hf}, \boldsymbol{C_{g|f}}\right)$, with p.d.f

$$p(\boldsymbol{g} \mid \boldsymbol{f}) \propto \exp\left(-\frac{1}{2}\|\boldsymbol{g} - \boldsymbol{Hf}\|^2_{\boldsymbol{C}^{-1}_{\boldsymbol{g}|f}}\right).$$

- The p.d.f of the prior is f such that,

$$p(\boldsymbol{f}) \propto \exp\left(-\frac{1}{2}\|\boldsymbol{f} - \overline{\boldsymbol{f}}\|^2_{C_f^{-1}}\right).$$

- The posterior is

$$p(\boldsymbol{f} \mid \boldsymbol{g}) \propto p(\boldsymbol{g} \mid \boldsymbol{f})p(\boldsymbol{f}).$$

- MAP solution $\hat{f}_{MAP}$:

$$\hat{\boldsymbol{f}}_{MAP} = \arg\max_{\boldsymbol{f}} p(\boldsymbol{f} \mid \boldsymbol{g}) = \arg\min_{\boldsymbol{f}}\{-\ln(p(\boldsymbol{g} \mid \boldsymbol{f}) - \ln(p(\boldsymbol{f}))\}$$

$$\begin{aligned}\hat{\boldsymbol{f}}_{MAP} &= \arg\min_{\boldsymbol{f}}\left(\|\boldsymbol{g} - \boldsymbol{Hf}\|^2_{C^{-1}_{g|f}} + \|\boldsymbol{f} - \overline{\boldsymbol{f}}\|^2_{C_f^{-1}}\right).\\ &= \arg\min_{\boldsymbol{f}} \Phi_{\boldsymbol{f}|\boldsymbol{g}}(\boldsymbol{f}).\end{aligned}$$

Computing the solution :

$$\begin{aligned}\Phi_{\boldsymbol{f}|\boldsymbol{g}}(\boldsymbol{f}) =& \langle \boldsymbol{g} - \boldsymbol{Hf}, \boldsymbol{g} - \boldsymbol{Hf}\rangle_{\boldsymbol{C}^{-1}_{\boldsymbol{g}|\boldsymbol{f}}} + \langle \boldsymbol{f} - \overline{\boldsymbol{f}}, \boldsymbol{f} - \overline{\boldsymbol{f}}\rangle_{\boldsymbol{C}_f^{-1}}\\ &= (\boldsymbol{g} - \boldsymbol{Hf})^T \boldsymbol{C}^{-1}_{\boldsymbol{g}|\boldsymbol{f}}(\boldsymbol{g} - \boldsymbol{Hf}) + (\boldsymbol{f} - \overline{\boldsymbol{f}})^T \boldsymbol{C}^{-1}_{\boldsymbol{f}}(\boldsymbol{f} - \overline{\boldsymbol{f}}).\\ \frac{\partial \Phi_{\boldsymbol{f}|\boldsymbol{g}}}{\partial \boldsymbol{f}} &= 0 \Leftrightarrow 2\boldsymbol{H}^T \boldsymbol{C}^{-1}_{\boldsymbol{g}|\boldsymbol{f}}(\boldsymbol{Hf} - \boldsymbol{g}) + 2\boldsymbol{C}^{-1}_{\boldsymbol{f}}(\boldsymbol{f} - \overline{\boldsymbol{f}}) = 0\\ \hat{\boldsymbol{f}}_{MAP} &= \left(\boldsymbol{C}^{-1}_{\boldsymbol{f}} + \boldsymbol{H}^T \boldsymbol{C}^{-1}_{\boldsymbol{g}|\boldsymbol{f}}\boldsymbol{H}\right)^{-1} \cdot \left(\boldsymbol{H}^T \boldsymbol{C}^{-1}_{\boldsymbol{g}|\boldsymbol{f}}\boldsymbol{g} + \boldsymbol{C}^{-1}_{\boldsymbol{f}}\overline{\boldsymbol{f}}\right).\end{aligned}$$

3.4.9 Bayesian approach

- How to choose the prior on f?
- How to choose the parameters?
- What if we don't have an explicit solution?
- Etc.
- $\Rightarrow$ Next lecture.

3.4.10 Exercise

Exercise 1

Prior model for the noise : $\boldsymbol{b} \sim \mathcal{N}\left(\mathbf{0}, \sigma_b^2 \boldsymbol{I}\right)$. Let $\gamma_{\boldsymbol{b}} = \frac{1}{\sigma_b^2}$, denote the precision.

1. Q1: Show that the prior density on b is:

$$p\left(\boldsymbol{b} \mid \gamma_b\right) = (2\pi)^{-N/2} \gamma_b^{N/2} \exp\left(-\frac{\gamma_b}{2}\|\boldsymbol{b}\|^2\right).$$

- Using the recall on the density of a multidimensional Gaussian, the density of b is :

$$p\left(\boldsymbol{b} \mid \gamma_b\right) = \frac{1}{(2\pi)^{N/2}\left|\boldsymbol{\Sigma}_b\right|^{1/2}} \exp\left[-\frac{1}{2}\boldsymbol{b}^T \boldsymbol{\Sigma}_b{}^{-1}\boldsymbol{b}\right].$$

- Since $\boldsymbol{\Sigma_b}$ is a diagonal matrix, $\Sigma_{\boldsymbol{b}} = \operatorname{diag}\left(\sigma_b^2, \ldots, \sigma_b^2\right)$, its determinant is :

$$\left|\boldsymbol{\Sigma}_b\right| = \prod_{i=1}^{N} \sigma_b^2 = \prod_{i=1}^{N} \frac{1}{\gamma_b} = \left(\gamma_b\right)^{-N}.$$

- Its inverse is $\boldsymbol{\Sigma}_b^{-1} = \operatorname{diag}\left(\frac{1}{\sigma_b^2}, \ldots, \frac{1}{\sigma_b^2}\right) \overset{i=1}{=} \operatorname{diag}\left(\gamma_b, \ldots, \gamma_b\right) = \gamma_b \boldsymbol{I}$. - We thus have

$$p\left(\boldsymbol{b} \mid \gamma_b\right) = (2\pi)^{-N/2} \gamma_b^{N/2} \exp[-\frac{\gamma_b}{2} \underbrace{\boldsymbol{b}^T\boldsymbol{b}}_{=\|\boldsymbol{b}\|^2}].$$

2. We have

   - Since the likelihood is given by:

$$p\left(\boldsymbol{g} \mid \boldsymbol{f}, \gamma_b\right) = (2\pi)^{-N/2} \gamma_b^{N/2} \exp\left(-\frac{\gamma_b}{2}\|\boldsymbol{g} - \boldsymbol{H}\boldsymbol{f}\|^2\right),$$

   - The log-likelihood is:

$$\ln\left[p\left(\boldsymbol{g} \mid \boldsymbol{f}, \gamma_b\right)\right] = -\frac{N}{2}\ln(2\pi) + \frac{N}{2}\ln\left(\gamma_b\right) - \frac{\gamma_b}{2}\|\boldsymbol{g} - \boldsymbol{H}\boldsymbol{f}\|^2.$$

   Deduce from that $\|\boldsymbol{g} - \boldsymbol{H}\boldsymbol{f}\|^2$.

$$\|\boldsymbol{g} - \boldsymbol{H}\boldsymbol{f}\|^2 = C_g - k_{\boldsymbol{g}} \ln\left[p\left(\boldsymbol{g} \mid \boldsymbol{f}, \gamma_b\right)\right]$$

where

$$C_g = -\frac{N}{\gamma_b}\ln(2\pi) + \frac{N}{\gamma_b}\ln\left(\gamma_b\right) \text{ and } k_g = \frac{2}{\gamma_b}.$$

3.
$$p\left(\boldsymbol{f} \mid \gamma_f\right) = (2\pi)^{-N/2}|\boldsymbol{\Pi}|^{1/2}\gamma_f^{N/2} \exp\left(-\frac{\gamma_f}{2}\boldsymbol{f}^T\boldsymbol{\Pi}\boldsymbol{f}\right).$$

   - Using the recall on the multidimensionnal distribution, the prior density is :

$$p\left(\boldsymbol{f} \mid \gamma_{\boldsymbol{f}}\right) = (2\pi)^{-N/2}|\boldsymbol{R}|^{-1/2} \exp\left[-\frac{1}{2}\boldsymbol{f}^T\boldsymbol{R}^{-1}\boldsymbol{f}\right].$$

- Since the determinant of the inverse R^{-1} is :

$$|\boldsymbol{R}|^{-1}=\frac{1}{|\boldsymbol{R}|} \Leftrightarrow |\boldsymbol{R}|=\frac{1}{|\boldsymbol{R}|^{-1}},$$

- Let's compute this determinant : $\left|\boldsymbol{R}^{-1}\right| = |\gamma_f \mathbf{\Pi}| = (\gamma_f)^N |\mathbf{\Pi}|$.
- This implies that:

$$|\boldsymbol{R}|=\frac{1}{\left|\boldsymbol{R}^{-1}\right|}=(\gamma_f)^{-N}|\mathbf{\Pi}|^{-1} \Rightarrow |\boldsymbol{R}|^{-1/2}=(\gamma_f)^{N/2}|\mathbf{\Pi}|^{1/2}.$$

- Putting this value of $|\boldsymbol{R}|^{-1/2}$ into equation (1), and replacing R^{-1} with its value, we have :

$$p(\boldsymbol{f} \mid \gamma_{\boldsymbol{f}})=(2\pi)^{-N/2}(\gamma_f)^{N/2}|\mathbf{\Pi}|^{1/2}\exp\left[-\frac{\gamma_f}{2}\boldsymbol{f}^T\mathbf{\Pi}\boldsymbol{f}\right].$$

4. By Q3, we have the log likelihood of $p(f \mid \gamma_f)$ is

$$\ln\left[p(f \mid \gamma_f)\right]=-\frac{N}{2}\ln(2\pi)+\frac{N}{2}\ln(\gamma_f)+\frac{1}{2}\ln(|\Pi|)-\frac{\gamma_f}{2}f^T\Pi f$$

Therefore, we can deduce that

$$\begin{aligned}\Phi_0(f):=f^T\Pi f&=\frac{1}{\gamma_f}\left(-N\ln(2\pi)+N\ln(\gamma_f)+\ln|\Pi|\right)-\frac{2}{\gamma_f}\ln\left[p(f \mid \gamma_f)\right]\\&=C_f-k_f\ln\left[p(f \mid \gamma_f)\right]\end{aligned}$$

5. The joint distribution of f and g is computed as

$$\begin{aligned}p(\boldsymbol{g},\boldsymbol{f} \mid \gamma_f,\gamma_b)&=p(\boldsymbol{g} \mid \boldsymbol{f},\gamma_b)\,p(\boldsymbol{f} \mid \gamma_f)\\&=(2\pi)^{-N/2}\gamma_b^{N/2}\exp\left(-\frac{\gamma_b}{2}\|\boldsymbol{g}-\boldsymbol{H}\boldsymbol{f}\|^2\right)\\&\times(2\pi)^{-N/2}\gamma_f^{N/2}|\mathbf{\Pi}|^{1/2}\exp\left[-\frac{\gamma_f}{2}\boldsymbol{f}^T\mathbf{\Pi}\boldsymbol{f}\right]\\&=(2\pi)^{-N}|\mathbf{\Pi}|^{1/2}\gamma_f^{N/2}\gamma_b^{N/2}\exp\left(-\frac{1}{2}\left[\gamma_b\|\boldsymbol{g}-\boldsymbol{H}\boldsymbol{f}\|^2+\gamma_f\|\boldsymbol{f}\|_{\mathbf{\Pi}}^2\right]\right).\end{aligned}$$

This joint distribution is parametrized by the two precision parameters γ_b and γ_f. We are going now to parameterize it using one parameter.

6. Factorizing by γ_b in the exponential, we have :

$$\begin{aligned}&\exp\left(-\frac{\gamma_b}{2}\left[\|\boldsymbol{g}-\boldsymbol{H}\boldsymbol{f}\|^2+\frac{\gamma_f}{\gamma_b}\|\boldsymbol{f}\|_{\mathbf{\Pi}}^2\right]\right)\\&=\exp\left(-\frac{\gamma_b}{2}\left[\|\boldsymbol{g}-\boldsymbol{H}\boldsymbol{f}\|^2+\mu\|\boldsymbol{f}\|_{\mathbf{\Pi}}^2\right]\right)\quad\text{where } \mu=\frac{\gamma_f}{\gamma_b}\\&=\exp\left(-\frac{\gamma_b}{2}\left[\Phi_{LS}(\boldsymbol{f})+\mu\Phi_0(\boldsymbol{f})\right]\right)\\&=\exp\left(-\frac{\gamma_b}{2}\Phi_{PLS}(\boldsymbol{f})\right)\end{aligned}$$

Comment: The regularisation parameter μ is written as the inverse of the signal-to-noise ratio : $\mu=\dfrac{\gamma_f}{\gamma_b}=\dfrac{\delta_b^\infty}{\sigma_f^2}$. $\hookrightarrow$ The higher the variance of the noise, the higher μ is, and the more we will favor the prior (regularize).

7. - We have

$$p(\boldsymbol{g}, \boldsymbol{f}, \gamma_{\boldsymbol{b}}, \gamma_f) = p(\boldsymbol{g}, \boldsymbol{f} \mid \gamma_{\boldsymbol{b}}, \gamma_f)\, p(\gamma_f)\, p(\gamma_b)\,.$$

   - Using the previous result on $p(\boldsymbol{g}, \boldsymbol{f} \mid \gamma_b, \gamma_f)$, and the prior density on γ_f and γ_b, we find after computation :

$$\begin{aligned} p(\boldsymbol{g}, \boldsymbol{f}, \gamma_b, \gamma_f) &= p(\boldsymbol{g}, \boldsymbol{f} \mid \gamma_b, \gamma_f)\, p(\gamma_f)\, p(\gamma_b) \\ &= (2\pi)^{-N} |\boldsymbol{\Pi}|^{1/2} \frac{\beta^{\alpha_b} \beta^{\alpha_f}}{\Gamma(\alpha_b)\,\Gamma(\alpha_f)} \gamma_f^{\alpha_f + N/2 - 1} \gamma_b^{\alpha_b + N/2 - 1} \\ &\times \exp\left[-\left[\gamma_b \left(\beta_b + \frac{\|\boldsymbol{g} - \boldsymbol{H}\boldsymbol{f}\|^2}{2}\right) + \gamma_f \left(\beta_f + \frac{\|\boldsymbol{f}\|^2_{\boldsymbol{\Pi}}}{2}\right)\right]\right] \end{aligned}$$

8. Deduce from this the joint posterior density (up to a constant) of the object of interest and the hyper-parameters, $p(\boldsymbol{f}, \gamma_b, \gamma_f \mid \boldsymbol{g})$.

$$\begin{aligned} p(\boldsymbol{f}, \gamma_b, \gamma_f \mid \boldsymbol{g}) &= \frac{p(\boldsymbol{f}, \boldsymbol{g}, \gamma_b, \gamma_f)}{p(\boldsymbol{g})} \\ &\propto \gamma_f^{\alpha_f + N/2 - 1} \gamma_b^{\alpha_b + N/2 - 1} \\ &\times \exp\left[-\left[\gamma_b \left(\beta_b + \frac{\|\boldsymbol{g} - \boldsymbol{H}\boldsymbol{f}\|^2}{2}\right) + \gamma_f \left(\beta_f + \frac{\|\boldsymbol{f}\|^2_{\Pi}}{2}\right)\right]\right]. \end{aligned}$$

9. Compute
   - FCD of γ_b.
     Picking up the terms that involve γ_b in the joint posterior density,

$$p(\gamma_b \mid \boldsymbol{f}, \gamma_f, \boldsymbol{g}) \propto \gamma_b^{\alpha_b + N/2 - 1} \exp\left[-\gamma_b \left(\beta_b + \frac{\|\boldsymbol{g} - \boldsymbol{H}\boldsymbol{f}\|^2}{2}\right)\right]$$

     - Then $\gamma_b \mid \boldsymbol{f}, \gamma_f, \boldsymbol{g} \sim \text{Gamma}\left(\alpha_b + \frac{N}{2}; \beta_b + \frac{\|\boldsymbol{g} - \boldsymbol{H}\boldsymbol{f}\|^2}{2}\right)$.
     Remark. - The second parameter of this Gamma distribution requires computing the norm of the modeling error: $\|\boldsymbol{g} - \boldsymbol{H}\boldsymbol{f}\|^2$.
     - The computational cost of this norm in the spatial domain is usually high.
     - Computing it in the Fourier domain reduces this cost.
   - FCD of γ_f
     - In the same way, picking up the terms that involve γ_f in the joint posterior density,

$$p(\gamma_f \mid \boldsymbol{f}, \gamma_b, \boldsymbol{g}) \propto \gamma_f^{\alpha_f + N/2 - 1} \exp\left[-\gamma_f \left(\beta_f + \frac{\|\boldsymbol{f}\|^2_{\Pi}}{2}\right)\right]$$

     - Then $\gamma_f \mid \boldsymbol{f}, \gamma_b, \boldsymbol{g} \sim \text{Gamma}\left(\alpha_f + \frac{N}{2}; \beta_f + \frac{\|\boldsymbol{f}\|^2_{\Pi}}{2}\right)$. - Remark. - As before, the second parameter of this Gamma distribution requires to compute the norm with respect to the regularization matrix. - For the same reasons as before, the passage in the Fourier domatin allows to reduce the computation cost.
   - FCD of f
     - Similarly,

$$\begin{aligned} p(\boldsymbol{f} \mid \gamma_b, \gamma_f, \boldsymbol{g}) &\propto \exp\left[-\left(\gamma_b \frac{\|\boldsymbol{g} - \boldsymbol{H}\boldsymbol{f}\|^2}{2} + \gamma_f \frac{\|\boldsymbol{f}\|^2_{\boldsymbol{\Pi}}}{2}\right)\right] \\ &= \exp\left[-\frac{\gamma_b}{2} \Phi_{PLS}(\boldsymbol{f})\right] \text{ where } \mu = \frac{\gamma_f}{\gamma_b}. \end{aligned}$$

- The density itself is Gaussian as the argument inside the exponential is a positive definite quadratic equation in $\boldsymbol{f}$.
- It is fully characterised by its mean $\mu_{f|...}$ and its covariance matrix $\boldsymbol{\Sigma}_{f|...}$.
- The mean $\mu_{f|....}$ is also the mode (maximum) and as such, also the minimizer of $\phi_{PLS}(\boldsymbol{f})$ (penalized least squares solution), i.e the Wiener-Hunt solution.
- The covariance matrix $\boldsymbol{\Sigma}_{f|...}$ is obtained by calculating the Hessian of $\Phi_{PLS}(f)$.

Remark 3.4.8

- The numerical problem one now faces is the sampling of a potentially high dimensional Gaussian distribution.
- This high dimensionality prohibits the inversion or factorisation of the covariance matrix $\boldsymbol{\Sigma}_{\boldsymbol{f}|...}$.
- What can we do to overcome this issue, ?
- To overcome this issue, we employ a circulant approximation of the covariance matrix to gain access to fast matrix operations performed in the Fourier domain. **Recall:**

$$\boldsymbol{H} = \boldsymbol{F}^* \Lambda_H \boldsymbol{F} \text{ and } \boldsymbol{H}^T = \boldsymbol{F}^* \Lambda_H^* \boldsymbol{F} \Rightarrow \boldsymbol{H}^T \boldsymbol{H} = \boldsymbol{F}^* \Lambda_H^* \Lambda_H \boldsymbol{F}.$$

- The matrix $\Lambda_H^* \Lambda_H$ being diagonal, this result means that $H^T H$ is diagonalisable in the Fourier basis and becomes $\Lambda_H^* \Lambda_H$ in the Fourier domain.
- In the same way, we show that $\boldsymbol{\Pi} = \boldsymbol{D}^T \boldsymbol{D}$ is diagonalizable in the Fourier basis and becomes in the Fourier domain $\Lambda_D^* \Lambda_D$.
- We then obtain the following expressions for the covariance matrix and the mean in the Fourier domain :

$$\begin{aligned} \boldsymbol{\Lambda}_{f|...} &= (\gamma_b \Lambda_{\boldsymbol{H}}^* \Lambda_{\boldsymbol{H}} + \gamma_f \Lambda_D^* \Lambda_D)^{-1} \\ \mathring{\boldsymbol{\mu}}_{f|...} &= \gamma_b \boldsymbol{\Lambda}_{\boldsymbol{f}|...} \Lambda_{\boldsymbol{H}}^* \, \mathring{\boldsymbol{g}} \end{aligned}$$

- In the Fourier domain, the covariance matrix is diagonal which means that its components are decorrelated. As such, each component is independent of the rest which enables one to sample them in parallel.